International Paper

RIVERDALE MILL
601 COUNTY ROAD 78
SELMA, AL 36703
PHONE: (334) 872-3481

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February 26, 2026

Mr. Aubrey H. White III
Chief, Air Division
Alabama Department of Environmental Management
1400 Coliseum Boulevard
Montgomery, Alabama 36110-2059

Subject: Condensate Treatment System
First Quarter 2026 – Compliance Verification Test

Dear Mr. White:

International Paper's Riverdale mill conducted a compliance verification test (CVT) on the condensate treatment system from January 27, 2026, through January 31, 2026. The results of the CVT are 11.1 pounds of hazardous air pollutants (HAPs) treated per oven dried ton of pulp production and 87.3% collected HAPs treated. The regulations require that at least 10.2 # HAPs/ODTP or at least 92% of collected HAPs be treated. Therefore, International Paper's Riverdale mill is in compliance with the condensate treatment portion of the Cluster Rule regulations.

The daily surrogate parameter during the test averaged 94.0 pounds soluble COD per horsepower. We will continue using 131 #sCOD/hp as the daily surrogate.

Please call me at 334/418-5236 if you have comments or questions.

Sincerely,

Carla C. Pearson
Environmental Services

Enclosures

INTERNATIONAL PAPER

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ADEM AIR DIVISION

FIRST QUARTER 2026
CONDENSATE TREATMENT

COMPLIANCE VERIFICATION TEST

For 40 CFR 63, Subpart S
(Pulp and Paper NESHAP)

CONDENSATE TREATMENT SYSTEM
Riverdale Mill

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1.0 NOTIFICATION OF COMPLIANCE STATUS

Based on information and belief formed after reasonable inquire, I certify in accordance with 40 CFR 63.9(h) that the statements and information in this document are true and accurate, and the source has complied with the standard as discussed below.

Micah Evans
Riverdale Mill Manager

Date 3/9/2026

1.1 Methods Used to Determine Compliance

The methods used to determine compliance include hazardous air pollutant and methanol analyses (i.e., NCASI Method DI/MeOH + HAPs 94.01 and DI/MeOH 94.03), temperature measurements, WATER9 computer model, total blower and aerator horsepower in treatment zone, flow monitoring devices, pulp production factors and others as described in the IPT Plan submitted May 7, 2007.

1.2 Methods Used for Continuing Compliance

During each quarter, the mill will verify compliance with the following parameters: at least 10.2 # HAPs/ODTP and/or at least 92% treatment of collected HAPs. Total HAPs will be determined annually and a methanol to HAPs ratio (r) will be developed for use during the year's remaining quarters.

Daily compliance will be verified using the surrogate ratio of 5 day rolling total soluble chemical oxygen demand to horsepower (sCOD/hp). This surrogate ratio will be verified and may be reset during any quarterly compliance verification test having 45 day ADEM notification. The surrogate ratio for this quarterly test was 94.0# sCOD/hp; therefore, we will continue to use 131# sCOD/hp as the surrogate ratio.

1.3 Major Source Determination

International Paper Riverdale mill is a major source requiring compliance with the Cluster Rule Regulations.

1.4 Statement of Compliance

The result of this condensate treatment compliance verification test (CVT) is 11.1 pounds of hazardous air pollutants (HAPs) treated per oven dried pulp production (i.e., # HAPs/ODTP) and 87.3% of the collected HAPs were treated. The regulations require that at least 10.2 # HAPs/ODTP or greater than 92% of collected HAPs be treated. Therefore, International Paper's Riverdale mill is in compliance with the condensate treatment portion of the Cluster Rule regulations.

2.0 INTRODUCTION

2.1 Summary of Test Program

International Paper's Riverdale mill conducted a compliance verification test (CVT) to demonstrate compliance with the condensate treatment requirements of CFR Part 63.446(e)(2). This regulation requires treatment of 92% or greater of the collected HAPs. The collection portion of the regulations require that 11.1 pounds of HAPs per oven-dried ton of pulp produced be collected. Therefore, assuming compliance with the collection portion of the rule, treatment of 92% of the collected HAPS would require that 10.2 # HAPs/ODTP be treated. This report provides the results of the IPT.

The CVT consisted of monitoring and sampling at: the main condensate collection tank outlet to the hard-pipe, inlets to the 5-Day and 60-Day aerated stabilization basins (ASBs), four zones within the ASB system, and the ASB system outlet. Pulp mill production data was collected during the days of testing to calculate oven dry tons of pulp (ODTP) produced. For more details on the sampling protocol and CMS justification, please refer to the Results of Initial Performance Test and CMS Performance Evaluation submitted to ADEM on May 7, 2007.

2.2 Test Results

International Paper's Riverdale mill conducted the five (5) days of sample collection for the CVT on January 27, 2026 through January 31, 2026, and followed the test plan submitted to ADEM in the above referenced document. The test results can be found in Appendix C.

3.0 SOURCE DESCRIPTION AND SAMPLING METHODOLOGY

3.1 Process Description

The Riverdale Mill produces paper from hardwood and softwood. Batch digesters cook the wood by the Kraft process. Pulp is "blown" from the digesters to three blow tanks and digester off-gasses are condensed in the turpentine and blow heat condenser.

The condensate streams collected to meet the mass collection requirement 40 CFR 63.446(c)(3) are transported in a closed collection system via a hardpipe per 63.446(d)(1) to the ASB.

3.2 Control Equipment Description

The total collected condensate streams are hard piped from the collection tank to the ASB system. This is an open biological treatment system that contains a fine bubble diffused aeration system with four 600 hp blowers and a surface aerated zone with twenty-one 75 hp aerators. One to four blowers and zero to twenty-one aerators normally operate dependent on waste load. During this testing, four blowers and an average of 24 aerators were operating.

The ASB system is divided into four well-mixed zones. The first zone begins with the inlet and the fourth zone ends at the end of the curtain divider. Basin characterization studies revealed uniform properties in each of these areas.

4.0 SUMMARY AND DISCUSSION OF RESULTS

4.1 Objectives and Test Matrix

The objective of the CVT was to demonstrate compliance with the 10.2 lb/ODTP condensate treatment option.

INITIAL PERFORMANCE TEST MATRIX

Sampling Location	Number of Samples per Run	Sample/Type of Pollutant	Sampling Method	Sampling Organization	Analytical Method	Analytical Laboratory
Total foul collected condensate	One grab sample per run. A performance test is 5 daily runs	Total HAPs – methanol, MEK, proprionaldehyde, and acetaldehyde concentration (mg/l)	Grab	Weston	NCASI DI/HAPS-99.01, NCASI DI/ MEOH94.03	Weston
Total collected foul condensate flow	Continuous data collection per day	Combined condensate flow (gal/day)	Continuous	Data Acquisition System	NA	NA
Total foul collected condensate	One grab sample per run	sCOD,	Grab	Weston	Method 410.4 using HACH COD reactor and HACH DR2010 analyzer	Mill Laboratory
Inlet to ASB	One grab sample per run	Methanol concentration (mg/l)	Grab	Weston	NCASI DI/ MEOH94.03	Weston
Inlet to ASB	One grab sample per run	sCOD,	Grab	Weston	Method 410.4 using HACH COD reactor and HACH DR2010 analyzer	Mill Laboratory
Inlet to ASB Flow	Continuous data collection per day	Mill Raw Water Flow (gal/day)	Continuous	Data Acquisition System	NA	NA
In-zone 1 through 4 samples	Grab samples per run	Methanol concentration (mg/l)	Grab	Weston	NCASI DI/ MEOH94.03	Weston
ASB Outlet	Grab samples per run	Methanol concentration (mg/l)	Grab	Weston	NCASI DI/ MEOH94.03	Weston
Digesters batch blow count	Continuous data collection per day	Mill production (ODTP/Day)/24 = ODTP/hr	Continuous	Data Acquisition System	NA	NA
Aeration	Total blowers and aerators running per run	Aeration	Visual count	Mill Personnel Documentation	NA	NA
Basin Temperature	One grab sample per zone per run.	Temperature	Grab	Weston	NA	Weston

4.2 Field Test Changes and Problems

No problems occurred and no changes were made to the submitted test plan.

4.3 Summary of Results for CMS Ranges

Appendix B contains a summary of compliance. Also shown is the continuing compliance surrogate parameter limit of 131 # sCOD/hp. This ratio is the maximum required for daily compliance (i.e., daily ratios below this are compliant) and may be reset with any compliance verification. The surrogate parameter during this test was 94.0# sCOD/hp; therefore, we will continue to use 131 # sCOD/hp as the surrogate parameter. Annual notification of quarterly compliance verifications with verbal notification the day of the test will be given to ADEM in lieu of the 15-day notification required in the rule. The annual quarterly testing schedule was submitted to ADEM in November, 2024.

5.0 Sampling and Analytical Procedures

5.1 Emission Test Methods

The test incorporated the appropriate QA/QC procedures specified in the NCASI Test Method 94.1, 94.03

5.2 Process Test Methods

The collected condensates flow meter and the effluent treatment flow meter were calibrated per the manufacturer's recommendations. Calibration reports are maintained in the mill's maintenance files.

5.3 Sample Identification and Custody

Mill personnel were responsible for collecting the samples and ensuring that all samples were accounted for and that proper custody procedures were followed. After collecting the samples, sample labels and integrity seals were applied and chain of custody forms were completed. Copies of all the COCs are included in Appendix B.

6.0 QA/QC Activities

Weston Solutions, Inc. and the International Paper Riverdale Lab have QA/QC plans that were followed for their analyses.

Appendix A

Calculations

Batch digester pulp production:

ODT/D= (SW Cooks * ODTSW/C) + (HW Cooks * ODTHW/C)

ODT/D = Oven Dried Tons of Pulp Produced per Day
SW Cooks = Number of batch digester softwood cooks per day
ODTSW/C = Finance factor for oven dried tons softwood produced per cook
HW Cooks = Number of batch digester hardwood cooks per day
ODTSW/C = Finance factor for oven dried tons hardwood produced per cook

Methanol collected:

F(b1) =E(bi)/P:

F(b1) = Mass flow rate of methanol in the condensate stream entering the treatment system for each run i (lb/ODTP)
E(b1) = Mass flow rate of methanol in the condensate stream entering the treatment system for each run i, (lb/hr).
P = Daily production (ODTP/D)

Total HAPs treated:

F = Fb * (f bio(MeOH)/(1 + 1.087(r)))

F = methanol mass removal (lb/ODTP) per each one hour run (adjusted for total HAPs)
Fb = inlet mass flow rate of methanol (lb/ODTP) determined using the procedures in paragraph above
fbio(MeOH) = the fraction of methanol removed in the biological treatment system.
r = ratio of the sum of acetaldehyde, methyl ethyl ketone, and propionaldehyde mass to methanol mass in the condensate stream entering the treatment system.

Continued Daily Compliance Calculations

The sCOD/hp will be documented for each one hour run from the stream entering the ASB for treatment. The average sCOD/hp for the three runs will establish the upper limit for continuous compliance indication. Multiple initial performance tests may be conducted or any valid performance test may be used to establish this upper limit. sCOD/hp is calculated using the following equation:

$$\text{sCOD/hp} = \left(\frac{[(\text{ASB Inlet Pounds sCOD/Day}) + (\text{Foul Condensate Pounds sCOD/Day})]}{[(600\ \text{hp/blower} * \#\ \text{blowers}) + (75\ \text{hp/aerator} * \#\ \text{aerators})]} \right)$$

Appendix B

Summary of Compliance
&
Form 13 Data

Compliance Verification Test Results for January 27, 2026 through January 31, 2026

Date	January 27, 2026
Total Production ODTUP/15 day	15,547
Methanol Mass in Hard Pipe (lb/15 day)	203,036
Methanol in condensate (lb/ODUTP):	13.1
Fraction of Methanol Biodegraded Fbio:	0.855
r-factor	0.0230
Total HAPs treated (lb/ODTUP):	**10.9**
sCOD/hp (5-day rolling avg)	**82**

Date	January 28, 2026
Total Production ODTUP/15 day	15,321
Methanol Mass in Hard Pipe (lb/15 day)	201,254
Methanol in condensate (lb/ODUTP):	13.1
Fraction of Methanol Biodegraded Fbio:	0.893
r-factor	0.0230
Total HAPs treated (lb/ODTUP):	**11.4**
sCOD/hp (5-day rolling avg)	**83**

Date	January 29, 2026
Total Production ODTUP/15 day	14,700
Methanol Mass in Hard Pipe (lb/15 day)	192,094
Methanol in condensate (lb/ODUTP):	13.1
Fraction of Methanol Biodegraded Fbio:	0.856
r-factor	0.0230
Total HAPs treated (lb/ODTUP):	**10.9**
sCOD/hp (5-day rolling avg)	**100**

Date	January 30, 2026
Total Production ODTUP/15 day	14,303
Methanol Mass in Hard Pipe (lb/15 day)	183,166
Methanol in condensate (lb/ODUTP):	12.8
Fraction of Methanol Biodegraded Fbio:	0.906
r-factor	0.0230
Total HAPs treated (lb/ODTUP):	**11.3**
sCOD/hp (5-day rolling avg)	**102**

Date	January 31, 2026
Total Production ODTUP/15 day	14,098
Methanol Mass in Hard Pipe (lb/15 day)	180,908
Methanol in condensate (lb/ODUTP):	12.8
Fraction of Methanol Biodegraded Fbio:	0.855
r-factor	0.0230
Total HAPs treated (lb/ODTUP):	**10.7**
sCOD/hp (5-day rolling avg)	**103**

January 27, 2026 through January 31, 2026	
Avg. Total Production	**14,794**
Avg. Methanol Mass in Hard Pipe	**192,092**
Avg. Methanol in condensate (lb/ODUTP):	**13.0**
Avg. Fraction of Methanol Biodegraded Fbio:	**0.873**
r-factor	**0.0230**
Avg. Total HAPs treated (lb/ODTUP):	**11.1**
Avg sCOD/hp (lb/hp)	**94.0**

5-day ASB Grid Sampling Data

Locations	Temperature		Methanol
1/27/2026	oF	oC	ppm
Hardpipe	48	9	3000.0
Inlet	98	37	38.6
Zone 1 - Point 2	98	37	44.7
Zone 2 - Point K	96	36	56.5
Zone 3 - Point R	94	34	54.4
Point 7	94	34	49.90
Zone 4 - Point 10	64	18	0.57
Tip of Curtain - Point 12	58	14	0.57

# Aerators - Zone 4	25
Hardpipe Flow (mgd)	0.297
5Dy Inlet Flow (mgd)	41.9
Daily sCOD/hp (lb/hp)	101
5Day COD/hp (lb/hp)	82

15 Day MeOH Mass Collected	203,036
15 Day (ODTUP/15 day)	15,547
Daily (ODTUP/day)	752.6

Additional Required Input Data

Date	1/27/2026
Annual Average Windspeed @ Maxwell AFB (mph)	5.4
Production (ODTUP/day)	753
15 Day Production (ODTUP/15 day)	15,547
15 Day MeOH Mass Collected (lb MeOH/15 day)	203,036
Inlet Stream Flow (mgd)	41.900
Inlet Stream MeOH (mg/L)	38.6
Hardpipe Stream Flow (mgd)	0.297
Hardpipe Stream MeOH (mg/L)	3,000
# of Equilivant Aerators 5-Day ASB*	
Zone 1	0
Zone 2 (max 8)	8
Zone 3 (max 24)	24
# of Aerators 60-Day ASB	
Zone 4	25
Daily sCOD/hp (lb/hp)	101
Surrogate Parameter sCOD/hp (lb/hp) 5-day rolling	82

Notes:
*1 blower = 8 aerators
*2 blowers = 16 aerators
*3 blowers = 24 aerators
Non-detect methanol values are entered as half the detection limit, based on NCASI and ADEM air guidance.
(current detection limit is 0.4 ppm, therefore non-detect = 0.2 ppm)

	ug/l		
Methanol	3244	16220	3244
Acetaldehyde	41.46	207.3	41.46
Propional	1.636	8.18	1.636
MEK	31.62	158.1	31.62

r Factor	0.0230

CCA CMS Four-Quarter Rolling Fbio & Volatilization Factor Average					
	1 QTR 25	2 QTR 25	3 QTR 25	4 QTR 25	AVG
5-day ASB - Average					
Percent Volatilized (%)	10.4	12.1	22.1	17.8	15.6
Percent Biodegraded (%)	12.5	1.5	-29.9	-29.8	-11.4
60-day ASB - Average					
Percent Volatilized (%)	0.1	0.1	0.3	0.2	0.2
Percent Biodegraded (%)	99.0	99.3	99.0	98.1	98.9

Date	January 27, 2026
Total Production ODTUP/15 day	15,547
Methanol Mass in Hard Pipe (lb/15 day)	203,036
Methanol in condensate (lb/ODUTP):	13.1
Fraction of Methanol Biodegraded Fbio:	0.855
r-factor	0.0230
Total HAPs treated (lb/ODTUP):	**10.9**
sCOD/hp (5-day rolling avg)	**82**

Volatilization & Biodegradation Values							
5-day ASB				**60-day ASB**			
Percent Volatilized		13.4	%	Percent Volatilized		0.2	%
Mass Volatilized		1.75	lb/ODTUP	Mass Volatilized		0.02	lb/ODTUP
Percent Biodegraded		2.7	%	Percent Biodegraded		98.7	%
Mass Biodegraded		0.35	lb/ODTUP	Mass Biodegraded		10.81	lb/ODTUP
Mass Breakthrough		10.96	lb/ODTUP	Mass Breakthrough		0.13	lb/ODTUP

Notes:
The r-factor is updated once a year in the first quarter.

Data Date: 1/27/26

I. BIOTREATMENT UNIT DESCRIPTION						
		5-Day ASB			60-Day ASB	
	Units	Zone 1	Zone 2	Zone 3	Zone 4	
Number of 75 HP Surface Aerators	#	0	8	24	25	
Total Horsepower	hp	0	600	1800	1875	
Temperature	°C	36.7	35.6	34.4	17.8	31.1
Length	ft	865	206	1674	2590	
Width	ft	432	413	558	1295	
Average Depth	ft	14	14	14	6	
Aerator Rotation	rpm	1200	1200	1200	1200	
Agitation Area per Aerator	ft^2	1452	1452	1452	1452	
Impeller Diameter	in	23.9	23.9	23.9	23.9	

II. OVERALL PARAMS - Individual Flows & MeOH Concentrations				
Parameter		Flow MGD	MeOH mg/L	MeOH lb/day
5-Day ASB Inlet**		41.90	38.6	13,489
Hard Pipe Inlet		0.30	3,000	7,435

***except condensate flow*

II. OVERALL PARAMS - Total Flows & Wind Speed			
	Flow m^3/sec	Flow MGD	MeOH mg/L
5-Day ASB Total	NA	NA	NA
Wind Speed	mph	5.44	

III. HAP DATA - Methanol & Zone Temperatures								
			5-Day ASB				60-Day ASB	Curtain tip west side
	Units	Inlet	Zone 1	Zone 2	Zone 3	Cross-over	Zone 4	
Conc.	mg/L	59.5	44.70	56.50	54.40	49.90	0.57	0.57
Temp.	°F		98.0	96.0	94.0	94	64.0	Detection Limit
								0.4

IV. RESULTS			
Partitioning Split by Ponds		5 Day ASB	60 Day ASB
Fraction biodegraded		2.7	98.7
Fraction air emissions		13.4	0.2
Fraction remaining in unit effluent		83.9	1.1

IV. RESULTS (Net Combined Efficiency of 5-Day & 60-Day ASBs		
Overall Partitioning		%
Fraction biodegraded		85.5
Fraction air emissions		13.5
Fraction remaining in unit effluent		1.0

I. ADDITIONAL BIOTREATMENT UNIT DESCRIPTION						
Active Volume						
5-Day ASB			60-Day ASB			
Zone 1	Zone 2	Zone 3	Zone 4		Totals	Units
148,157	33,732	370,349	569,920		1,122,157	m^3
39.1	8.9	97.8	150.6		296.4	MG
Surface Area						
5-Day ASB			60-Day ASB			
Zone 1	Zone 2	Zone 3	Zone 4		Totals	
373,680	85,078	934,092	3,354,050		4,746,900	ft^2
34,716	7,904	86,780	311,601		441,001	m^2
8.6	2.0	21.4	77.0		109.0	acres

Parameters For Calculating Mass Transfer Coefficients
International Paper - Riverdale Mill

Data Date: 01/27/26

HAP Compound	Diff in Water	Diff in Air	Henry's Law	Equil. Ratio (Hc) or (Keq)	MW	ScG	Antoine Eqtn	
	cm2/s	cm2/s	atm-m3/mol	m3 liq to m3 gas	g/mol		b	c
Methanol	1.64E-05	0.150	6.46E-06	2.64E-04	32.0	1.006	1474	229.13
Acetaldehyde	1.41E-05	0.124	8.77E-05	3.58E-03	45.1	1.216	1600	291.8
MEK	9.80E-06	0.081	1.30E-04	5.31E-03	72.1	1.867	1305	229.27

General Parameters & Constants

PARAMETER	Units	Value	Name
viscosity of air	g/cm-s	0.000181	va
viscosity of water	g/cm-s	0.002	vw
density of air	g/cm3	0.0012	da
density of water	g/cm3	1	dw
MW of air	g/mol	29	Mwa
MW of water	g/mol	18	MWw
Diff of O2 in H2O	cm2/s	2.40E-05	DO2w
grav const.	lb-ft/s2/lb	32.17	g
R	atm-m3/mol K	8.21E-05	R
Aerator Motor Eff	fraction	0.85	AerEff
O2 Trans Correct		0.83	Beta
Wind Speed	m/s	2.43	U
Diff of Ether	m/s	8.50E-06	Dether
O2 Trans	lb O2/HP-h	3	J

These Parameters are used when F/D < 14 AND U > 3.25 m/s

TURBULENT

KL Params	5-Day ASB Zone 1	5-Day ASB Zone 2	5-Day ASB Zone 3	60-Day ASB Zone 4
w	126.3	126.3	126.3	126.3
Re	3.11E+06	3.11E+06	3.11E+06	3.11E+06
PI	0	35,063	35,063	35,063
Power Number, p	0.00E+00	2.86E-04	2.86E-04	2.86E-04
Fr	9.88E+02	9.88E+02	9.88E+02	9.88E+02
Total TurbArea (ft2)	0	3872	11616	36300
Frac. Agitated	0.000	0.046	0.012	0.011
(by surface aerators)				
QUIESCENT				
Depth	4.27	4.27	4.27	1.83
SurfArea (ft2)	373,680	85,078	934,092	3,354,050
SurfArea (m2)	34,762	7,914	86,894	312,011
F/D Ratio	49	23	78	344
ScL - Methanol	NA	NA	NA	NA
ScL - Acetaldehyde	NA	NA	NA	NA
ScL - MEK	NA	NA	NA	NA
U* (Friction Velocity)	NA	NA	NA	NA
DIFFUSED				
Air flow, cfm	0	12,500	12,500	0
Air flow, m3/s	0.000	5.899	5.899	0.000

Calculating Mass Transfer Coefficient KL for Various Zones
International Paper - Riverdale Mill

Data Date: **01/27/26**

	Surface Aeration												
	Turbulent Area					Quiescient Area KL, m/s							OVERALL
ZONE	Temp Adj H	kG m/s	kL m/s	KL turb m/s		kG m/s	U10 < 3.25	U10 > 3.25 F/D<14	14<F/D<51.2	F/D>51.2	kL m/s	KL quisc m/s	KL overall m/s
5-Day ASB: Zone1													
Methanol	1.16E-05	0.00E+00	0.00E+00	0.00E+00		5.33E-03	4.31E-06	NA	2.35E-06	2.39E-06	4.31E-06	1.56E-06	1.56E-06
5-Day ASB: Zone2													
Methanol	1.10E-05	1.44E-01	6.82E-02	6.25E-05		5.78E-03	4.31E-06	NA	1.73E-06	2.39E-06	4.31E-06	1.59E-06	4.36E-06
5-Day ASB: Zone3													
Methanol	1.04E-05	1.44E-01	6.65E-02	5.95E-05		5.07E-03	4.31E-06	NA	1.17E-06	2.39E-06	4.31E-06	1.41E-06	2.13E-06
60-Day ASB: Zone4													
Methanol	4.37E-06	1.44E-01	1.49E-02	2.64E-05		4.73E-03	4.31E-06	NA	9.39E-06	2.39E-06	4.31E-06	7.21E-07	9.99E-07

	Submerged Aeration - KD		
	Temp. Adj. Hc (Keq)	Equiv. KL m/s	KL overall
5-Day ASB: Zone1			
Methanol	4.74E-04	0.00E+00	1.56E-06
5-Day ASB: Zone2			
Methanol	4.50E-04	3.35E-07	4.70E-06
5-Day ASB: Zone3			
Methanol	4.26E-04	2.89E-08	2.16E-06

FORM XIII. DATA FORM FOR THE ESTIMATION OF MULTIPLE ZONE BIODEGRADATION FROM UNIT CONCENTRATIONS

Data Date: 01/27/26

NAME OF THE FACILITY	**IP - RIVERDALE**	
COMPOUND for site specific biorate determination	**MeOH**	
Number of zones in the biological treatment unit	1	
VOLUME of full-scale system (cubic meters)	2	718076.8224
Average DEPTH of the full-scale system (meters)	3	3.66
FLOW RATE of wastewater to the unit (m3/s)	4	1.836
FLOW RATE of condensate to the unit (m3/s)	5	0.013
Total wastewater flowrate - (including condensates) (m3/s)	5-A	NA
ESTIMATE OF KL (m/s)	6	*see table*
Concentration in the wastewater treated in the unit (mg/L)	7	38.6
Concentration in the condensates (mg/L)	8	3,000
Concentration in wastewater (total - inc. cond) in (mg/L)	8-A	NA
Concentration in the treated effluent (mg/L)	9	0.57
TOTAL INLET FLOW (m3/s) line 4 plus the number on line 5 (or 5-A)	10	1.849
TOTAL RESIDENCE TIME (s) line 2 divided by line 10.	11	388453
TOTAL AREA OF IMPOUNDMENT (m2) line 2 divided by line 3	12	196196

	Lines 13 through 15 Not Used						**Overall**
				Estimate of KL			
Zone	**Concentration for zone,**		**Area of the**	**in the zone**		**AIR STRIPPING**	
Number	**Ci (mg/L)**		**zone, A (m2)**	**(m/s)**		**KL A Ci (g/s)**	
5-Day ASB Z-1	44.70		34714.87	1.56E-06			2.41
5-Day ASB Z-2	56.50		7903.75	4.70E-06			2.10
5-Day ASB Z-3	54.40		86777.15	2.16E-06			10.20
60-Day ASB-Z4	0.57		311591.25	9.99E-07			0.18
TOTALS - sum for each zone.		**15**	**440987.01**		**16**		**14.89**

Removal by air stripping (g/s). Line 16.	17	14.89
Loading in effluent (g/s). Line 9 times line 10.	18	1.05
Total loading (g/s). {(line 5*line 8)+(line 4*line 7)} or {line 5-A*line 8-A}.	19	109.9
Removal by biodegradation (g/s) Line 19 minus (line 17 + line 18).	20	94.0
Fraction biodegraded: Divide line 20 by line 19.	21	0.855
Fraction air emissions: Divide line 17 by line 19.	22	0.135
Fraction remaining in unit effluent. Divide line 18 by 19.	23	0.010

Date	January 27, 2026
Total Production ODTUP/15 day	15,547
Methanol Mass in Hard Pipe (lb/15 day)	203,036
Methanol in condensate (lb/ODUTP):	13.1
Fraction of Methanol Biodegraded Fbio:	0.855
r-factor	0.0230
Total HAPs treated (lb/ODTUP):	**10.9**
sCOD/hp (5-day rolling avg)	**82**

Averages

Fraction of Methanol Biodegraded Fbio:	**0.855**
5-day average Total HAPS Treated	**10.89**
sCOD/hp (lb/hp)	**82**

DATA FORM FOR THE ESTIMATION OF THE HENRY'S LAW CONSTANT FOR A COMPOUND IN THE BIOLOGICAL TREATMENT UNIT			
Form IX			
NAME OF THE FACILITY for site specific biorate determination	Riverdale Mill		
COMPOUND for site specific biorate determination			Methanol
LISTED HENRY'S LAW VALUE AT 25 degrees Celsius. (ratio of mol fraction in gas to mole fraction in water at one atmosphere)	1		0.2885
TEMPERATURE of the liquid in the unit (deg.C)	2		31.111111
CALCULATION OF K			
Temperature adjusted Henry's law value (equals the value on line 1 if the temperature on line 2 is 25)	3		0.3590
Temperature in degrees Kelvin. Add 273.16 to the number on line 2. Enter the results here.	4		304.3
Temperature ratio. Divide 273.16 by the number on line 4. Enter the results here.	5		0.8978
Henry's Law adjustment factor. Multiply the number on line 5 by 0.804 and enter the results here.	6		0.7218
Henry's Law value (g/m3 gas per g/m3 liquid) Multiply the number on line 3 by the number on line 6 and divide the results by 1000. Enter the results here.	7		2.59E-04
Henry's Law value (atm m3 per mol) Divide the number on line 3 by 55555 and enter the results here.	8		6.462E-06

5-day ASB Grid Sampling Data

Locations	Temperature		Methanol
1/28/2026	oF	oC	ppm
Hardpipe	68	20	3270.0
Inlet	88	31	34.5
Zone 1 - Point 2	96	36	32.6
Zone 2 - Point K	96	36	41.4
Zone 3 - Point R	96	36	43.3
Point 7	96	36	42.30
Zone 4 - Point 10	60	16	0.57
Tip of Curtain - Point 12	60	16	0.57

# Aerators - Zone 4	25
Hardpipe Flow (mgd)	0.415
5Dy Inlet Flow (mgd)	39.1
Daily sCOD/hp (lb/hp)	95
5Day COD/hp (lb/hp)	83

15 Day MeOH Mass Collected	201,254
15 Day (ODTUP/15 day)	15,321
Daily (ODTUP/day)	746.5

Additional Required Input Data

Date	1/28/2026
Annual Average Windspeed @ Maxwell AFB (mph)	5.4
Production (ODTUP/day)	747
15 Day Production (ODTUP/15 day)	15,321
15 Day MeOH Mass Collected (lb MeOH/15 day)	201,254
Inlet Stream Flow (mgd)	39.060
Inlet Stream MeOH (mg/L)	34.5
Hardpipe Stream Flow (mgd)	0.415
Hardpipe Stream MeOH (mg/L)	3,270
# of Equilivant Aerators 5-Day ASB*	
Zone 1	0
Zone 2 (max 8)	8
Zone 3 (max 24)	24
# of Aerators 60-Day ASB	
Zone 4	25
Daily sCOD/hp (lb/hp)	95
Surrogate Parameter sCOD/hp (lb/hp) 5-day rolling	83

Notes:
*1 blower = 8 aerators
*2 blowers = 16 aerators
*3 blowers = 24 aerators
Non-detect methanol values are entered as half the detection limit, based on NCASI and ADEM air guidance.
(current detection limit is 0.4 ppm, therefore non-detect = 0.2 ppm)

	ug/l		
Methanol	3244	16220	3244
Acetaldehyde	41.46	207.3	41.46
Propional	1.636	8.18	1.636
MEK	31.62	158.1	31.62

r Factor	0.0230

CCA CMS Four-Quarter Rolling Fbio & Volatilization Factor Average					
	1 QTR 25	2 QTR 25	3 QTR 25	4 QTR 25	AVG
5-day ASB - Average					
Percent Volatilized (%)	10.4	12.1	22.1	17.8	15.6
Percent Biodegraded (%)	12.5	1.5	-29.9	-29.8	-11.4
60-day ASB - Average					
Percent Volatilized (%)	0.1	0.1	0.3	0.2	0.2
Percent Biodegraded (%)	99.0	99.3	99.0	98.1	98.8

Date	January 28, 2026
Total Production ODTUP/15 day	15,321
Methanol Mass in Hard Pipe (lb/15 day)	201,254
Methanol in condensate (lb/ODUTP):	13.1
Fraction of Methanol Biodegraded Fbio:	0.893
r-factor	0.0230
Total HAPs treated (lb/ODTUP):	11.4
sCOD/hp (5-day rolling avg)	83

Volatilization & Biodegradation Values							
5-day ASB				**60-day ASB**			
Percent Volatilized		9.7	%	Percent Volatilized		0.2	%
Mass Volatilized		1.28	lb/ODTUP	Mass Volatilized		0.02	lb/ODTUP
Percent Biodegraded		28.5	%	Percent Biodegraded		98.4	%
Mass Biodegraded		3.74	lb/ODTUP	Mass Biodegraded		7.99	lb/ODTUP
Mass Breakthrough		8.11	lb/ODTUP	Mass Breakthrough		0.11	lb/ODTUP

Notes:
The r-factor is updated once a year in the first quarter.

Input Data
International Paper - Riverdale Mill

Data Date: 1/28/26

I. BIOTREATMENT UNIT DESCRIPTION						
		5-Day ASB			60-Day ASB	
	Units	Zone 1	Zone 2	Zone 3	Zone 4	
Number of 75 HP Surface Aerators	#	0	8	24	25	
Total Horsepower	hp	0	600	1800	1875	
Temperature	°C	35.6	35.6	35.6	15.6	30.6
Length	ft	865	206	1674	2590	
Width	ft	432	413	558	1295	
Average Depth	ft	14	14	14	6	
Aerator Rotation	rpm	1200	1200	1200	1200	
Agitation Area per Aerator	ft^2	1452	1452	1452	1452	
Impeller Diameter	in	23.9	23.9	23.9	23.9	

II. OVERALL PARAMS - Individual Flows & MeOH Concentrations				
		Flow	MeOH	MeOH
Parameter		MGD	mg/L	lb/day
5-Day ASB Inlet**		39.06	34.5	11,239
Hard Pipe Inlet		0.41	3,270	11,309

***except condensate flow*

II. OVERALL PARAMS - Total Flows & Wind Speed			
	Flow m^3/sec	Flow MGD	MeOH mg/L
5-Day ASB Total	NA	NA	NA
Wind Speed	mph	5.44	

III. HAP DATA - Methanol & Zone Temperatures								
			5-Day ASB			Cross-over	60-Day ASB	Curtain tip west side
	Units	Inlet	Zone 1	Zone 2	Zone 3		Zone 4	
Conc.	mg/L	68.5	32.60	41.40	43.30	42.30	0.57	0.57
Temp.	°F		96.0	96.0	96.0	96	60.0	Detection Limit
								0.4

IV. RESULTS			
Partitioning Split by Ponds		5 Day ASB	60 Day ASB
Fraction biodegraded		28.5	98.4
Fraction air emissions		9.7	0.2
Fraction remaining in unit effluent		61.8	1.3

IV. RESULTS (Net Combined Efficiency of 5-Day & 60-Day ASBs		
Overall Partitioning		%
Fraction biodegraded		89.3
Fraction air emissions		9.9
Fraction remaining in unit effluent		0.8

I. ADDITIONAL BIOTREATMENT UNIT DESCRIPTION						
Active Volume						
5-Day ASB			60-Day ASB			
Zone 1	Zone 2	Zone 3	Zone 4		Totals	Units
148,157	33,732	370,349	569,920		1,122,157	m^3
39.1	8.9	97.8	150.6		296.4	MG
Surface Area						
5-Day ASB			60-Day ASB			
Zone 1	Zone 2	Zone 3	Zone 4		Totals	
373,680	85,078	934,092	3,354,050		4,746,900	ft^2
34,716	7,904	86,780	311,601		441,001	m^2
8.6	2.0	21.4	77.0		109.0	acres

Parameters For Calculating Mass Transfer Coefficients
International Paper - Riverdale Mill

Data Date: **01/28/26**

HAP Compound	Diff in Water	Diff in Air	Henry's Law	Equil. Ratio (Hc) or (Keq)	MW	ScG	Antoine Eqtn	
	cm2/s	cm2/s	atm-m3/mol	m3 liq to m3 gas	g/mol		b	c
Methanol	1.64E-05	0.150	6.35E-06	2.59E-04	32.0	1.006	1474	229.13
Acetaldehyde	1.41E-05	0.124	8.77E-05	3.58E-03	45.1	1.216	1600	291.8
MEK	9.80E-06	0.081	1.30E-04	5.31E-03	72.1	1.867	1305	229.27

General Parameters & Constants

PARAMETER	Units	Value	Name
viscosity of air	g/cm-s	0.000181	va
viscosity of water	g/cm-s	0.002	vw
density of air	g/cm3	0.0012	da
density of water	g/cm3	1	dw
MW of air	g/mol	29	Mwa
MW of water	g/mol	18	MWw
Diff of O2 in H2O	cm2/s	2.40E-05	DO2w
grav const.	lb-ft/s2/lb	32.17	g
R	atm-m3/mol K	8.21E-05	R
Aerator Motor Eff	fraction	0.85	AerEff
O2 Trans Correct		0.83	Beta
Wind Speed	m/s	2.43	U
Diff of Ether	m/s	8.50E-06	Dether
O2 Trans	lb O2/HP-h	3	J

These Parameters are used when F/D < 14 AND U > 3.25 m/s

TURBULENT

	5-Day ASB	5-Day ASB	5-Day ASB	60-Day ASB
KL Params	Zone 1	Zone 2	Zone 3	Zone 4
w	126.3	126.3	126.3	126.3
Re	3.11E+06	3.11E+06	3.11E+06	3.11E+06
PI	0	35,063	35,063	35,063
Power Number, p	0.00E+00	2.86E-04	2.86E-04	2.86E-04
Fr	9.88E+02	9.88E+02	9.88E+02	9.88E+02
Total TurbArea (ft2)	0	3872	11616	36300
Frac. Agitated	0.000	0.046	0.012	0.011
(by surface aerators)				
QUIESCENT				
Depth	4.27	4.27	4.27	1.83
SurfArea (ft2)	373,680	85,078	934,092	3,354,050
SurfArea (m2)	34,762	7,914	86,894	312,011
F/D Ratio	49	23	78	344
ScL - Methanol	NA	NA	NA	NA
ScL - Acetaldehyde	NA	NA	NA	NA
ScL - MEK	NA	NA	NA	NA
U* (Friction Velocity)	NA	NA	NA	NA
DIFFUSED				
Air flow, cfm	0	12,500	12,500	0
Air flow, m3/s	0.000	5.899	5.899	0.000

Calculating Mass Transfer Coefficient KL for Various Zones
International Paper - Riverdale Mill

Data Date: **01/28/26**

	Surface Aeration												
	Turbulent Area					Quiescient Area KL, m/s							OVERALL
ZONE	Temp Adj	kG	kL	KL turb		kG		U10 > 3.25			kL	KL quisc	KL overall
	H	m/s	m/s	m/s		m/s	U10 < 3.25	F/D<14	14<F/D<51.2	F/D>51.2	m/s	m/s	m/s
5-Day ASB: Zone1													
Methanol	1.08E-05	0.00E+00	0.00E+00	0.00E+00		5.33E-03	4.31E-06	NA	2.35E-06	2.39E-06	4.31E-06	1.49E-06	1.49E-06
5-Day ASB: Zone2													
Methanol	1.08E-05	1.44E-01	6.82E-02	6.14E-05		5.78E-03	4.31E-06	NA	1.73E-06	2.39E-06	4.31E-06	1.57E-06	4.29E-06
5-Day ASB: Zone3													
Methanol	1.08E-05	1.44E-01	6.82E-02	6.14E-05		5.07E-03	4.31E-06	NA	1.17E-06	2.39E-06	4.31E-06	1.44E-06	2.19E-06
60-Day ASB: Zone4													
Methanol	3.79E-06	1.44E-01	1.42E-02	2.30E-05		4.73E-03	4.31E-06	NA	9.39E-06	2.39E-06	4.31E-06	6.44E-07	8.86E-07

	Submerged Aeration - KD		
	Temp. Adj.	Equiv. KL	KL overall
	Hc (Keq)	m/s	
5-Day ASB: Zone1			
Methanol	4.42E-04	0.00E+00	1.49E-06
5-Day ASB: Zone2			
Methanol	4.42E-04	3.29E-07	4.62E-06
5-Day ASB: Zone3			
Methanol	4.42E-04	3.00E-08	2.22E-06

FORM XIII. DATA FORM FOR THE ESTIMATION OF MULTIPLE ZONE BIODEGRADATION FROM UNIT CONCENTRATIONS

Data Date: 01/28/26

NAME OF THE FACILITY	**IP - RIVERDALE**	
COMPOUND for site specific biorate determination	**MeOH**	
Number of zones in the biological treatment unit	1	
VOLUME of full-scale system (cubic meters)	2	718076.8224
Average DEPTH of the full-scale system (meters)	3	3.66
FLOW RATE of wastewater to the unit (m3/s)	4	1.711
FLOW RATE of condensate to the unit (m3/s)	5	0.018
Total wastewater flowrate - (including condensates) (m3/s)	5-A	NA
ESTIMATE OF KL (m/s)	6	*see table*
Concentration in the wastewater treated in the unit (mg/L)	7	34.5
Concentration in the condensates (mg/L)	8	3,270
Concentration in wastewater (total - inc. cond) in (mg/L)	8-A	NA
Concentration in the treated effluent (mg/L)	9	0.57
TOTAL INLET FLOW (m3/s) line 4 plus the number on line 5 (or 5-A)	10	1.729
TOTAL RESIDENCE TIME (s) line 2 divided by line 10.	11	415240
TOTAL AREA OF IMPOUNDMENT (m2) line 2 divided by line 3	12	196196

	Lines 13 through 15 Not Used						**Overall**
				Estimate of KL			
Zone	**Concentration for zone,**		**Area of the**	**in the zone**		**AIR STRIPPING**	
Number	**Ci (mg/L)**		**zone, A (m2)**	**(m/s)**		**KL A Ci (g/s)**	
5-Day ASB Z-1	32.60		34714.87	1.49E-06			1.68
5-Day ASB Z-2	41.40		7903.75	4.62E-06			1.51
5-Day ASB Z-3	43.30		86777.15	2.22E-06			8.33
60-Day ASB-Z4	0.57		311591.25	8.86E-07			0.16
TOTALS - sum for each zone.		**15**	**440987.01**		**16**		**11.68**

Removal by air stripping (g/s). Line 16.	17	11.68
Loading in effluent (g/s). Line 9 times line 10.	18	0.99
Total loading (g/s). {(line 5*line 8)+(line 4*line 7)} or {line 5-A*line 8-A}.	19	118.4
Removal by biodegradation (g/s) Line 19 minus (line 17 + line 18).	20	105.8
Fraction biodegraded: Divide line 20 by line 19.	21	0.893
Fraction air emissions: Divide line 17 by line 19.	22	0.099
Fraction remaining in unit effluent. Divide line 18 by 19.	23	0.008

Date	January 28, 2026
Total Production ODTUP/15 day	15,321
Methanol Mass in Hard Pipe (lb/15 day)	201,254
Methanol in condensate (lb/ODUTP):	13.1
Fraction of Methanol Biodegraded Fbio:	0.893
r-factor	0.0230
Total HAPs treated (lb/ODTUP):	**11.4**
sCOD/hp (5-day rolling avg)	**83**

Averages

Fraction of Methanol Biodegraded Fbio:	**#REF!**
5-day average Total HAPS Treated	**#REF!**
sCOD/hp (lb/hp)	**#REF!**

DATA FORM FOR THE ESTIMATION OF THE HENRY'S LAW CONSTANT FOR A COMPOUND IN THE BIOLOGICAL TREATMENT UNIT			
Form IX			
NAME OF THE FACILITY for site specific biorate determination	Riverdale Mill		
COMPOUND for site specific biorate determination			Methanol
LISTED HENRY'S LAW VALUE AT 25 degrees Celsius. (ratio of mol fraction in gas to mole fraction in water at one atmosphere)	1		0.2885
TEMPERATURE of the liquid in the unit (deg.C)	2		30.555556
CALCULATION OF K			
Temperature adjusted Henry's law value (equals the value on line 1 if the temperature on line 2 is 25)	3		0.3526
Temperature in degrees Kelvin. Add 273.16 to the number on line 2. Enter the results here.	4		303.7
Temperature ratio. Divide 273.16 by the number on line 4. Enter the results here.	5		0.8994
Henry's Law adjustment factor. Multiply the number on line 5 by 0.804 and enter the results here.	6		0.7231
Henry's Law value (g/m3 gas per g/m3 liquid) Multiply the number on line 3 by the number on line 6 and divide the results by 1000. Enter the results here.	7		2.55E-04
Henry's Law value (atm m3 per mol) Divide the number on line 3 by 55555 and enter the results here.	8		6.347E-06

5-day ASB Grid Sampling Data

Locations	Temperature		Methanol
1/29/2026	oF	oC	ppm
Hardpipe	90	32	3480.0
Inlet	90	32	28.6
Zone 1 - Point 2	92	33	49.1
Zone 2 - Point K	92	33	42.7
Zone 3 - Point R	94	34	38.5
Point 7	92	33	39.30
Zone 4 - Point 10	58	14	0.57
Tip of Curtain - Point 12	62	17	0.57

# Aerators - Zone 4	23
Hardpipe Flow (mgd)	0.230
5Dy Inlet Flow (mgd)	36.8
Daily sCOD/hp (lb/hp)	152
5Day COD/hp (lb/hp)	100

15 Day MeOH Mass Collected	192,094
15 Day (ODTUP/15 day)	14,700
Daily (ODTUP/day)	576.1

Additional Required Input Data

Date	1/29/2026
Annual Average Windspeed @ Maxwell AFB (mph)	5.4
Production (ODTUP/day)	576
15 Day Production (ODTUP/15 day)	14,700
15 Day MeOH Mass Collected (lb MeOH/15 day)	192,094
Inlet Stream Flow (mgd)	36.752
Inlet Stream MeOH (mg/L)	28.6
Hardpipe Stream Flow (mgd)	0.230
Hardpipe Stream MeOH (mg/L)	3,480
# of Equilivant Aerators 5-Day ASB*	
Zone 1	0
Zone 2 (max 8)	8
Zone 3 (max 24)	26
# of Aerators 60-Day ASB	
Zone 4	23
Daily sCOD/hp (lb/hp)	152
Surrogate Parameter sCOD/hp (lb/hp) 5-day rolling	100

Notes:
*1 blower = 8 aerators
*2 blowers = 16 aerators
*3 blowers = 24 aerators
Non-detect methanol values are entered as half the detection limit, based on NCASI and ADEM air guidance.
(current detection limit is 0.4 ppm, therefore non-detect = 0.2 ppm)

	ug/l		
Methanol	3244	16220	3244
Acetaldehyde	41.46	207.3	41.46
Propional	1.636	8.18	1.636
MEK	31.62	158.1	31.62

r Factor	0.0230

COMPLIANCE SUMMARY

CCA CMS Four-Quarter Rolling Fbio & Volatilization Factor Average					
	1 QTR 25	2 QTR 25	3 QTR 25	4 QTR 25	AVG
5-day ASB - Average					
Percent Volatilized (%)	10.4	12.1	22.1	17.8	15.6
Percent Biodegraded (%)	12.5	1.5	-29.9	-29.8	-11.4
60-day ASB - Average					
Percent Volatilized (%)	0.1	0.1	0.3	0.2	0.2
Percent Biodegraded (%)	99.0	99.3	99.0	98.1	98.9

Date	January 29, 2026
Total Production ODTUP/15 day	14,700
Methanol Mass in Hard Pipe (lb/15 day)	192,094
Methanol in condensate (lb/ODUTP):	13.1
Fraction of Methanol Biodegraded Fbio:	0.856
r-factor	0.0230
Total HAPs treated (lb/ODTUP):	**10.9**
sCOD/hp (5-day rolling avg)	**100**

Volatilization & Biodegradation Values							
5-day ASB				**60-day ASB**			
Percent Volatilized		13.1	%	Percent Volatilized		0.2	%
Mass Volatilized		1.71	lb/ODTUP	Mass Volatilized		0.02	lb/ODTUP
Percent Biodegraded		8.3	%	Percent Biodegraded		98.3	%
Mass Biodegraded		1.09	lb/ODTUP	Mass Biodegraded		10.10	lb/ODTUP
Mass Breakthrough		10.27	lb/ODTUP	Mass Breakthrough		0.15	lb/ODTUP

Notes:
The r-factor is updated once a year in the first quarter.

Data Date: 1/29/26

I. BIOTREATMENT UNIT DESCRIPTION						
		5-Day ASB			60-Day ASB	
	Units	Zone 1	Zone 2	Zone 3	Zone 4	
Number of 75 HP Surface Aerators	#	0	8	26	23	
Total Horsepower	hp	0	600	1950	1725	
Temperature	°C	33.3	33.3	34.4	14.4	28.9
Length	ft	865	206	1674	2590	
Width	ft	432	413	558	1295	
Average Depth	ft	14	14	14	6	
Aerator Rotation	rpm	1200	1200	1200	1200	
Agitation Area per Aerator	ft^2	1452	1452	1452	1452	
Impeller Diameter	in	23.9	23.9	23.9	23.9	

II. OVERALL PARAMS - Individual Flows & MeOH Concentrations				
Parameter		Flow MGD	/MeOH mg/L	MeOH lb/day
5-Day ASB Inlet**		36.75	28.6	8,766
Hard Pipe Inlet		0.23	3,480	6,662

***except condensate flow*

II. OVERALL PARAMS - Total Flows & Wind Speed			
	Flow m^3/sec	Flow MGD	MeOH mg/L
5-Day ASB Total	NA	NA	NA
Wind Speed	mph	5.44	

III. HAP DATA - Methanol & Zone Temperatures								
			5-Day ASB				60-Day ASB	Curtain tip west side
	Units	Inlet	Zone 1	Zone 2	Zone 3	Cross-over	Zone 4	
Conc.	mg/L	50.0	49.10	42.70	38.50	39.30	0.57	0.57
Temp.	°F		92.0	92.0	94.0	92	58.0	Detection Limit
								0.4

IV. RESULTS			
Partitioning Split by Ponds		5 Day ASB	60 Day ASB
Fraction biodegraded		8.3	98.3
Fraction air emissions		13.1	0.2
Fraction remaining in unit effluent		78.6	1.5

IV. RESULTS (Net Combined Efficiency of 5-Day & 60-Day ASBs		
Overall Partitioning		%
Fraction biodegraded		85.6
Fraction air emissions		13.3
Fraction remaining in unit effluent		1.1

I. ADDITIONAL BIOTREATMENT UNIT DESCRIPTION						
Active Volume						
5-Day ASB			60-Day ASB			
Zone 1	Zone 2	Zone 3	Zone 4		Totals	Units
148,157	33,732	370,349	569,920		1,122,157	m^3
39.1	8.9	97.8	150.6		296.4	MG
Surface Area						
5-Day ASB			60-Day ASB			
Zone 1	Zone 2	Zone 3	Zone 4		Totals	
373,680	85,078	934,092	3,354,050		4,746,900	ft^2
34,716	7,904	86,780	311,601		441,001	m^2
8.6	2.0	21.4	77.0		109.0	acres

Parameters For Calculating Mass Transfer Coefficients
International Paper - Riverdale Mill

Data Date: **01/29/26**

HAP Compound	Diff in Water cm2/s	Diff in Air cm2/s	Henry's Law atm-m3/mol	Equil. Ratio (Hc) or (Keq) m3 liq to m3 gas	MW g/mol	ScG	Antoine Eqtn b	Antoine Eqtn c
Methanol	1.64E-05	0.150	6.00E-06	2.45E-04	32.0	1.006	1474	229.13
Acetaldehyde	1.41E-05	0.124	8.77E-05	3.58E-03	45.1	1.216	1600	291.8
MEK	9.80E-06	0.081	1.30E-04	5.31E-03	72.1	1.867	1305	229.27

General Parameters & Constants

PARAMETER	Units	Value	Name
viscosity of air	g/cm-s	0.000181	va
viscosity of water	g/cm-s	0.002	vw
density of air	g/cm3	0.0012	da
density of water	g/cm3	1	dw
MW of air	g/mol	29	Mwa
MW of water	g/mol	18	MWw
Diff of O2 in H2O	cm2/s	2.40E-05	DO2w
grav const.	lb-ft/s2/lb	32.17	g
R	atm-m3/mol K	8.21E-05	R
Aerator Motor Eff	fraction	0.85	AerEff
O2 Trans Correct		0.83	Beta
Wind Speed	m/s	2.43	U
Diff of Ether	m/s	8.50E-06	Dether
O2 Trans	lb O2/HP-h	3	J

These Parameters are used when F/D < 14 AND U > 3.25 m/s

TURBULENT

KL Params	5-Day ASB Zone 1	5-Day ASB Zone 2	5-Day ASB Zone 3	60-Day ASB Zone 4
w	126.3	126.3	126.3	126.3
Re	3.11E+06	3.11E+06	3.11E+06	3.11E+06
PI	0	35,063	35,063	35,063
Power Number, p	0.00E+00	2.86E-04	2.86E-04	2.86E-04
Fr	9.88E+02	9.88E+02	9.88E+02	9.88E+02
Total TurbArea (ft2)	0	3872	12584	33396
Frac. Agitated	0.000	0.046	0.013	0.010
(by surface aerators)				
QUIESCENT				
Depth	4.27	4.27	4.27	1.83
SurfArea (ft2)	373,680	85,078	934,092	3,354,050
SurfArea (m2)	34,762	7,914	86,894	312,011
F/D Ratio	49	23	78	344
ScL - Methanol	NA	NA	NA	NA
ScL - Acetaldehyde	NA	NA	NA	NA
ScL - MEK	NA	NA	NA	NA
U* (Friction Velocity)	NA	NA	NA	NA
DIFFUSED				
Air flow, cfm	0	12,500	12,500	0
Air flow, m3/s	0.000	5.899	5.899	0.000

Calculating Mass Transfer Coefficient KL for Various Zones
International Paper - Riverdale Mill

Data Date: **01/29/26**

	Surface Aeration												
	Turbulent Area					Quiescient Area KL, m/s							OVERALL
ZONE	Temp Adj	kG	kL	KL turb		kG		U10 > 3.25			kL	KL quisc	KL overall
	H	m/s	m/s	m/s		m/s	U10 < 3.25	F/D<14	14<F/D<51.2	F/D>51.2	m/s	m/s	m/s
5-Day ASB: Zone1													
Methanol	9.17E-06	0.00E+00	0.00E+00	0.00E+00		5.33E-03	4.31E-06	NA	2.35E-06	2.39E-06	4.31E-06	1.34E-06	1.34E-06
5-Day ASB: Zone2													
Methanol	9.17E-06	1.44E-01	6.47E-02	5.25E-05		5.78E-03	4.31E-06	NA	1.73E-06	2.39E-06	4.31E-06	1.42E-06	3.74E-06
5-Day ASB: Zone3													
Methanol	9.68E-06	1.44E-01	6.65E-02	5.52E-05		5.07E-03	4.31E-06	NA	1.17E-06	2.39E-06	4.31E-06	1.34E-06	2.07E-06
60-Day ASB: Zone4													
Methanol	3.37E-06	1.44E-01	1.38E-02	2.05E-05		4.73E-03	4.31E-06	NA	9.39E-06	2.39E-06	4.31E-06	5.83E-07	7.82E-07

	Submerged Aeration - KD		
	Temp. Adj.	Equiv. KL	KL overall
	Hc (Keq)	m/s	
5-Day ASB: Zone1			
Methanol	3.75E-04	0.00E+00	1.34E-06
5-Day ASB: Zone2			
Methanol	3.75E-04	2.79E-07	4.02E-06
5-Day ASB: Zone3			
Methanol	3.96E-04	2.69E-08	2.09E-06

FORM XIII. DATA FORM FOR THE ESTIMATION OF MULTIPLE ZONE BIODEGRADATION FROM UNIT CONCENTRATIONS

Data Date: **01/29/26**

NAME OF THE FACILITY	**IP - RIVERDALE**	
COMPOUND for site specific biorate determination	**MeOH**	
Number of zones in the biological treatment unit	1	
VOLUME of full-scale system (cubic meters)	2	718076.8224
Average DEPTH of the full-scale system (meters)	3	3.66
FLOW RATE of wastewater to the unit (m3/s)	4	1.610
FLOW RATE of condensate to the unit (m3/s)	5	0.010
Total wastewater flowrate - (including condensates) (m3/s)	5-A	NA
ESTIMATE OF KL (m/s)	6	*see table*
Concentration in the wastewater treated in the unit (mg/L)	7	28.6
Concentration in the condensates (mg/L)	8	3,480
Concentration in wastewater (total - inc. cond) in (mg/L)	8-A	NA
Concentration in the treated effluent (mg/L)	9	0.57
TOTAL INLET FLOW (m3/s) line 4 plus the number on line 5 (or 5-A)	10	1.620
TOTAL RESIDENCE TIME (s) line 2 divided by line 10.	11	443231
TOTAL AREA OF IMPOUNDMENT (m2) line 2 divided by line 3	12	196196

	Lines 13 through 15 Not Used						**Overall**
				Estimate of KL			
Zone Number	**Concentration for zone, Ci (mg/L)**		**Area of the zone, A (m2)**	**in the zone (m/s)**		**AIR STRIPPING KL A Ci (g/s)**	
5-Day ASB Z-1	49.10		34714.87	1.34E-06			2.28
5-Day ASB Z-2	42.70		7903.75	4.02E-06			1.36
5-Day ASB Z-3	38.50		86777.15	2.09E-06			6.99
60-Day ASB-Z4	0.57		311591.25	7.82E-07			0.14
TOTALS - sum for each zone.		**15**	**440987.01**		**16**		**10.77**

Removal by air stripping (g/s). Line 16.	17	10.77
Loading in effluent (g/s). Line 9 times line 10.	18	0.92
Total loading (g/s). {(line 5*line 8)+(line 4*line 7)} or {line 5-A*line 8-A}.	19	81.0
Removal by biodegradation (g/s) Line 19 minus (line 17 + line 18).	20	69.3
Fraction biodegraded: Divide line 20 by line 19.	21	0.856
Fraction air emissions: Divide line 17 by line 19.	22	0.133
Fraction remaining in unit effluent. Divide line 18 by 19.	23	0.011

Date	January 29, 2026
Total Production ODTUP/15 day	14,700
Methanol Mass in Hard Pipe (lb/15 day)	192,094
Methanol in condensate (lb/ODUTP):	13.1
Fraction of Methanol Biodegraded Fbio:	0.856
r-factor	0.0230
Total HAPs treated (lb/ODTUP):	**10.9**
sCOD/hp (5-day rolling avg)	**100**

Averages

Fraction of Methanol Biodegraded Fbio:	**#REF!**
5-day average Total HAPS Treated	**#REF!**
sCOD/hp (lb/hp)	**#REF!**

DATA FORM FOR THE ESTIMATION OF THE HENRY'S LAW CONSTANT FOR A COMPOUND IN THE BIOLOGICAL TREATMENT UNIT			
Form IX			
NAME OF THE FACILITY for site specific biorate determination	Riverdale Mill		
COMPOUND for site specific biorate determination			Methanol
LISTED HENRY'S LAW VALUE AT 25 degrees Celsius. (ratio of mol fraction in gas to mole fraction in water at one atmosphere)	1		0.2885
TEMPERATURE of the liquid in the unit (deg.C)	2		28.888889
CALCULATION OF K			
Temperature adjusted Henry's law value (equals the value on line 1 if the temperature on line 2 is 25)	3		0.3334
Temperature in degrees Kelvin. Add 273.16 to the number on line 2. Enter the results here.	4		302.0
Temperature ratio. Divide 273.16 by the number on line 4. Enter the results here.	5		0.9044
Henry's Law adjustment factor. Multiply the number on line 5 by 0.804 and enter the results here.	6		0.7271
Henry's Law value (g/m3 gas per g/m3 liquid) Multiply the number on line 3 by the number on line 6 and divide the results by 1000. Enter the results here.	7		2.42E-04
Henry's Law value (atm m3 per mol) Divide the number on line 3 by 55555 and enter the results here.	8		6.001E-06

International Paper - Riverdale Mill
Input Data
Sample Collection Date: 1/31/2026
Mill Date: 1/30/2026

5-day ASB Grid Sampling Data

Locations	Temperature		Methanol
1/30/2026	oF	oC	ppm
Hardpipe	76	24	3140.0
Inlet	88	31	45.7
Zone 1 - Point 2	94	34	47.6
Zone 2 - Point K	94	34	44.6
Zone 3 - Point R	92	33	44.3
Point 7	86	30	39.2
Zone 4 - Point 10	54	12	0.6
Tip of Curtain - Point 12	62	17	0.6

# Aerators - Zone 4	23
Hardpipe Flow (mgd)	0.385
5Dy Inlet Flow (mgd)	40.0
Daily sCOD/hp (lb/hp)	82
5Day COD/hp (lb/hp)	102

15 Day MeOH Mass Collected	183,166
15 Day (ODTUP/15 day)	14,303
Daily (ODTUP/day)	904.8

Additional Required Input Data

Date	1/30/2026
Annual Average Windspeed @ Maxwell AFB (mph)	5.4
Production (ODTUP/day)	905
15 Day Production (ODTUP/15 day)	14,303
15 Day MeOH Mass Collected (lb MeOH/15 day)	183,166
Inlet Stream Flow (mgd)	39.965
Inlet Stream MeOH (mg/L)	45.7
Hardpipe Stream Flow (mgd)	0.385
Hardpipe Stream MeOH (mg/L)	3,140
# of Equilivant Aerators 5-Day ASB*	
Zone 1	0
Zone 2 (max 8)	8
Zone 3 (max 24)	26
# of Aerators 60-Day ASB	
Zone 4	23
Daily sCOD/hp (lb/hp)	82
Surrogate Parameter sCOD/hp (lb/hp) 5-day rolling	102

Notes:
*1 blower = 8 aerators
*2 blowers = 16 aerators
*3 blowers = 24 aerators
Non-detect methanol values are entered as half the detection limit, based on NCASI and ADEM air guidance.
(current detection limit is 0.4 ppm, therefore non-detect = 0.2 ppm)

	ug/l		
Methanol	3244	16220	3244
Acetaldehyde	41.46	207.3	41.46
Propional	1.636	8.18	1.636
MEK	31.62	158.1	31.62

r Factor	0.0230

CCA CMS Four-Quarter Rolling Fbio & Volatilization Factor Average					
	1 QTR 25	2 QTR 25	3 QTR 25	4 QTR 25	AVG
5-day ASB - Average					
Percent Volatilized (%)	10.4	12.1	22.1	17.8	15.6
Percent Biodegraded (%)	12.5	1.5	-29.9	-29.8	-11.4
60-day ASB - Average					
Percent Volatilized (%)	0.1	0.1	0.3	0.2	0.2
Percent Biodegraded (%)	99.0	99.3	99.0	98.1	98.9

Date	January 30, 2026
Total Production ODTUP/15 day	14,303
Methanol Mass in Hard Pipe (lb/15 day)	183,166
Methanol in condensate (lb/ODUTP):	12.8
Fraction of Methanol Biodegraded Fbio:	0.906
r-factor	0.0230
Total HAPs treated (lb/ODTUP):	**11.3**
sCOD/hp (5-day rolling avg)	**102**

Volatilization & Biodegradation Values							
5-day ASB				**60-day ASB**			
Percent Volatilized		8.6	%	Percent Volatilized		0.2	%
Mass Volatilized		1.10	lb/ODTUP	Mass Volatilized		0.01	lb/ODTUP
Percent Biodegraded		39.3	%	Percent Biodegraded		98.4	%
Mass Biodegraded		5.03	lb/ODTUP	Mass Biodegraded		6.57	lb/ODTUP
Mass Breakthrough	--	6.68	lb/ODTUP	Mass Breakthrough		0.10	lb/ODTUP

Notes:
The r-factor is updated once a year in the first quarter.

Data Date: 1/30/26

I. BIOTREATMENT UNIT DESCRIPTION		5-Day ASB			60-Day ASB
	Units	Zone 1	Zone 2	Zone 3	Zone 4
Number of 75 HP Surface Aerators	#	0	8	26	23
Total Horsepower	hp	0	600	1950	1725
Temperature	°C	34.4	34.4	33.3	12.2
Length	ft	865	206	1674	2590
Width	ft	432	413	558	1295
Average Depth	ft	14	14	14	6
Aerator Rotation	rpm	1200	1200	1200	1200
Agitation Area per Aerator	ft^2	1452	1452	1452	1452
Impeller Diameter	in	23.9	23.9	23.9	23.9

28.6

II. OVERALL PARAMS - Individual Flows & MeOH Concentrations				
Parameter		Flow MGD	MeOH mg/L	MeOH lb/day
5-Day ASB Inlet**		39.97	45.7	15,232
Hard Pipe Inlet		0.38	3,140	10,073

***except condensate flow*

II. OVERALL PARAMS - Total Flows & Wind Speed	Flow m^3/sec	Flow MGD	MeOH mg/L
5-Day ASB Total	NA	NA	NA
Wind Speed	mph	5.44	

III. HAP DATA - Methanol & Zone Temperatures				5-Day ASB			Cross-over	60-Day ASB	Curtain tip west side
		Units	Inlet	Zone 1	Zone 2	Zone 3		Zone 4	
	Conc.	mg/L	75.2	47.60	44.60	44.30	39.20	0.57	0.57
	Temp.	°F		94.0	94.0	92.0	86	54.0	Detection Limit
									0.4

IV. RESULTS		5 Day ASB	60 Day ASB
Partitioning Split by Ponds			
Fraction biodegraded		39.3	98.4
Fraction air emissions		8.6	0.2
Fraction remaining in unit effluent		52.1	1.5

IV. RESULTS (Net Combined Efficiency of 5-Day & 60-Day ASBs		%
Overall Partitioning		
Fraction biodegraded		90.6
Fraction air emissions		8.7
Fraction remaining in unit effluent		0.8

I. ADDITIONAL BIOTREATMENT UNIT DESCRIPTION						
Active Volume						
5-Day ASB			60-Day ASB			
Zone 1	Zone 2	Zone 3	Zone 4		Totals	Units
148,157	33,732	370,349	569,920		1,122,157	m^3
39.1	8.9	97.8	150.6		296.4	MG
Surface Area						
5-Day ASB			60-Day ASB			
Zone 1	Zone 2	Zone 3	Zone 4		Totals	
373,680	85,078	934,092	3,354,050		4,746,900	ft^2
34,716	7,904	86,780	311,601		441,001	m^2
8.6	2.0	21.4	77.0		109.0	acres

Parameters For Calculating Mass Transfer Coefficients
International Paper - Riverdale Mill

Data Date: 01/30/26

HAP Compound	Diff in Water cm2/s	Diff in Air cm2/s	Henry's Law atm-m3/mol	Equil. Ratio (Hc) or (Keq) m3 liq to m3 gas	MW g/mol	ScG	Antoine Eqtn b	Antoine Eqtn c
Methanol	1.64E-05	0.150	5.94E-06	2.43E-04	32.0	1.006	1474	229.13
Acetaldehyde	1.41E-05	0.124	8.77E-05	3.58E-03	45.1	1.216	1600	291.8
MEK	9.80E-06	0.081	1.30E-04	5.31E-03	72.1	1.867	1305	229.27

General Parameters & Constants

PARAMETER	Units	Value	Name
viscosity of air	g/cm-s	0.000181	va
viscosity of water	g/cm-s	0.002	vw
density of air	g/cm3	0.0012	da
density of water	g/cm3	1	dw
MW of air	g/mol	29	Mwa
MW of water	g/mol	18	MWw
Diff of O2 in H2O	cm2/s	2.40E-05	DO2w
grav const.	lb-ft/s2/lb	32.17	g
R	atm-m3/mol K	8.21E-05	R
Aerator Motor Eff	fraction	0.85	AerEff
O2 Trans Correct		0.83	Beta
Wind Speed	m/s	2.43	U
Diff of Ether	m/s	8.50E-06	Dether
O2 Trans	lb O2/HP-h	3	J

These Parameters are used when F/D < 14 AND U > 3.25 m/s

TURBULENT

KL Params	5-Day ASB Zone 1	5-Day ASB Zone 2	5-Day ASB Zone 3	60-Day ASB Zone 4
w	126.3	126.3	126.3	126.3
Re	3.11E+06	3.11E+06	3.11E+06	3.11E+06
PI	0	35,063	35,063	35,063
Power Number, p	0.00E+00	2.86E-04	2.86E-04	2.86E-04
Fr	9.88E+02	9.88E+02	9.88E+02	9.88E+02
Total TurbArea (ft2)	0	3872	12584	33396
Frac. Agitated	0.000	0.046	0.013	0.010
(by surface aerators)				
QUIESCENT				
Depth	4.27	4.27	4.27	1.83
SurfArea (ft2)	373,680	85,078	934,092	3,354,050
SurfArea (m2)	34,762	7,914	86,894	312,011
F/D Ratio	49	23	78	344
ScL - Methanol	NA	NA	NA	NA
ScL - Acetaldehyde	NA	NA	NA	NA
ScL - MEK	NA	NA	NA	NA
U* (Friction Velocity)	NA	NA	NA	NA
DIFFUSED				
Air flow, cfm	0	12,500	12,500	0
Air flow, m3/s	0.000	5.899	5.899	0.000

Calculating Mass Transfer Coefficient KL for Various Zones
International Paper - Riverdale Mill

Data Date: **01/30/26**

	Surface Aeration												
	Turbulent Area					Quiescient Area KL, m/s							OVERALL
ZONE	Temp Adj	kG	kL	KL turb		kG		U10 > 3.25			kL	KL quisc	KL overall
	H	m/s	m/s	m/s		m/s	U10 < 3.25	F/D<14	14<F/D<51.2	F/D>51.2	m/s	m/s	m/s
5-Day ASB: Zone1													
Methanol	9.59E-06	0.00E+00	0.00E+00	0.00E+00		5.33E-03	4.31E-06	NA	2.35E-06	2.39E-06	4.31E-06	1.38E-06	1.38E-06
5-Day ASB: Zone2													
Methanol	9.59E-06	1.44E-01	6.65E-02	5.47E-05		5.78E-03	4.31E-06	NA	1.73E-06	2.39E-06	4.31E-06	1.46E-06	3.88E-06
5-Day ASB: Zone3													
Methanol	9.08E-06	1.44E-01	6.47E-02	5.20E-05		5.07E-03	4.31E-06	NA	1.17E-06	2.39E-06	4.31E-06	1.28E-06	1.97E-06
60-Day ASB: Zone4													
Methanol	2.93E-06	1.44E-01	1.31E-02	1.80E-05		4.73E-03	4.31E-06	NA	9.39E-06	2.39E-06	4.31E-06	5.20E-07	6.95E-07

	Submerged Aeration - KD		
	Temp. Adj.	Equiv. KL	KL overall
	Hc (Keq)	m/s	
5-Day ASB: Zone1			
Methanol	3.92E-04	0.00E+00	1.38E-06
5-Day ASB: Zone2			
Methanol	3.92E-04	2.92E-07	4.17E-06
5-Day ASB: Zone3			
Methanol	3.71E-04	2.52E-08	1.99E-06

FORM XIII. DATA FORM FOR THE ESTIMATION OF MULTIPLE ZONE BIODEGRADATION FROM UNIT CONCENTRATIONS

Data Date: 01/30/26

NAME OF THE FACILITY	**IP - RIVERDALE**	
COMPOUND for site specific biorate determination	**MeOH**	
Number of zones in the biological treatment unit	1	
VOLUME of full-scale system (cubic meters)	2	718076.8224
Average DEPTH of the full-scale system (meters)	3	3.66
FLOW RATE of wastewater to the unit (m3/s)	4	1.751
FLOW RATE of condensate to the unit (m3/s)	5	0.017
Total wastewater flowrate - (including condensates) (m3/s)	5-A	NA
ESTIMATE OF KL (m/s)	6	*see table*
Concentration in the wastewater treated in the unit (mg/L)	7	45.7
Concentration in the condensates (mg/L)	8	3,140
Concentration in wastewater (total - inc. cond) in (mg/L)	8-A	NA
Concentration in the treated effluent (mg/L)	9	0.57
TOTAL INLET FLOW (m3/s) line 4 plus the number on line 5 (or 5-A)	10	1.768
TOTAL RESIDENCE TIME (s) line 2 divided by line 10.	11	406232
TOTAL AREA OF IMPOUNDMENT (m2) line 2 divided by line 3	12	196196

	Lines 13 through 15 Not Used						**Overall**
				Estimate of KL			
Zone Number	**Concentration for zone, Ci (mg/L)**		**Area of the zone, A (m2)**	**in the zone (m/s)**		**AIR STRIPPING KL A Ci (g/s)**	
5-Day ASB Z-1	47.60		34714.87	1.38E-06			2.28
5-Day ASB Z-2	44.60		7903.75	4.17E-06			1.47
5-Day ASB Z-3	44.30		86777.15	1.99E-06			7.66
60-Day ASB-Z4	0.57		311591.25	6.95E-07			0.12
TOTALS - sum for each zone.		**15**	**440987.01**		**16**		**11.53**

Removal by air stripping (g/s). Line 16.	17	11.53
Loading in effluent (g/s). Line 9 times line 10.	18	1.01
Total loading (g/s). {(line 5*line 8)+(line 4*line 7)} or {line 5-A*line 8-A}.	19	132.9
Removal by biodegradation (g/s) Line 19 minus (line 17 + line 18).	20	120.4
Fraction biodegraded: Divide line 20 by line 19.	21	0.906
Fraction air emissions: Divide line 17 by line 19.	22	0.087
Fraction remaining in unit effluent. Divide line 18 by 19.	23	0.008

Date	January 30, 2026
Total Production ODTUP/15 day	14,303
Methanol Mass in Hard Pipe (lb/15 day)	183,166
Methanol in condensate (lb/ODUTP):	12.8
Fraction of Methanol Biodegraded Fbio:	0.906
r-factor	0.0230
Total HAPs treated (lb/ODTUP):	**11.3**
sCOD/hp (5-day rolling avg)	**102**

Averages

Fraction of Methanol Biodegraded Fbio:	**#REF!**
5-day average Total HAPS Treated	**#REF!**
sCOD/hp (lb/hp)	**#REF!**

DATA FORM FOR THE ESTIMATION OF THE HENRY'S LAW CONSTANT FOR A COMPOUND IN THE BIOLOGICAL TREATMENT UNIT			
Form IX			
NAME OF THE FACILITY for site specific biorate determination	Riverdale Mill		
COMPOUND for site specific biorate determination			Methanol
LISTED HENRY'S LAW VALUE AT 25 degrees Celsius. (ratio of mol fraction in gas to mole fraction in water at one atmosphere)	1		0.2885
TEMPERATURE of the liquid in the unit (deg.C)	2		28.611111
CALCULATION OF K			
Temperature adjusted Henry's law value (equals the value on line 1 if the temperature on line 2 is 25)	3		0.3302
Temperature in degrees Kelvin. Add 273.16 to the number on line 2. Enter the results here.	4		301.8
Temperature ratio. Divide 273.16 by the number on line 4. Enter the results here.	5		0.9052
Henry's Law adjustment factor. Multiply the number on line 5 by 0.804 and enter the results here.	6		0.7278
Henry's Law value (g/m3 gas per g/m3 liquid) Multiply the number on line 3 by the number on line 6 and divide the results by 1000. Enter the results here.	7		2.40E-04
Henry's Law value (atm m3 per mol) Divide the number on line 3 by 55555 and enter the results here.	8		5.943E-06

International Paper - Riverdale Mill
Input Data
Sample Collection Date: 2/1/2026
Mill Date: 1/31/2026

5-day ASB Grid Sampling Data

Locations	Temperature		Methanol
1/31/2026	oF	oC	ppm
Hardpipe	88	31	3330.0
Inlet	76	24	10.7
Zone 1 - Point 2	92	33	33.8
Zone 2 - Point K	88	31	45.7
Zone 3 - Point R	92	33	43.2
Point 7	92	33	41.1
Zone 4 - Point 10	60	16	0.6
Tip of Curtain - Point 12	62	17	0.6

# Aerators - Zone 4	23
Hardpipe Flow (mgd)	0.414
5Dy Inlet Flow (mgd)	40.5
Daily sCOD/hp (lb/hp)	85
5Day COD/hp (lb/hp)	103

15 Day MeOH Mass Collected	180,908
15 Day (ODTUP/15 day)	14,098
Daily (ODTUP/day)	891.6

Additional Required Input Data

Date	1/31/2026
Annual Average Windspeed @ Maxwell AFB (mph)	5.4
Production (ODTUP/day)	892
15 Day Production (ODTUP/15 day)	14,098
15 Day MeOH Mass Collected (lb MeOH/15 day)	180,908
Inlet Stream Flow (mgd)	40.501
Inlet Stream MeOH (mg/L)	10.7
Hardpipe Stream Flow (mgd)	0.414
Hardpipe Stream MeOH (mg/L)	3,330
# of Equilivant Aerators 5-Day ASB*	
Zone 1	0
Zone 2 (max 8)	8
Zone 3 (max 24)	27
# of Aerators 60-Day ASB	
Zone 4	23
Daily sCOD/hp (lb/hp)	85
Surrogate Parameter sCOD/hp (lb/hp) 5-day rolling	103

Notes:
*1 blower = 8 aerators
*2 blowers = 16 aerators
*3 blowers = 24 aerators
Non-detect methanol values are entered as half the detection limit, based on NCASI and ADEM air guidance.
(current detection limit is 0.4 ppm, therefore non-detect = 0.2 ppm)

	ug/l		
Methanol	3244	16220	3244
Acetaldehyde	41.46	207.3	41.46
Propional	1.636	8.18	1.636
MEK	31.62	158.1	31.62

r Factor	0.0230

CCA CMS Four-Quarter Rolling Fbio & Volatilization Factor Average					
	1 QTR 25	2 QTR 25	3 QTR 25	4 QTR 25	AVG
5-day ASB - Average					
Percent Volatilized (%)	10.4	12.1	22.1	17.8	15.6
Percent Biodegraded (%)	12.5	1.5	-29.9	-29.8	-11.4
60-day ASB - Average					
Percent Volatilized (%)	0.1	0.1	0.3	0.2	0.2
Percent Biodegraded (%)	99.0	99.3	99.0	98.1	98.9

Date	January 31, 2026
Total Production ODTUP/15 day	14,098
Methanol Mass in Hard Pipe (lb/15 day)	180,908
Methanol in condensate (lb/ODUTP):	12.8
Fraction of Methanol Biodegraded Fbio:	0.855
r-factor	0.0230
Total HAPs treated (lb/ODTUP):	**10.7**
sCOD/hp (5-day rolling avg)	**103**

Volatilization & Biodegradation Values							
5-day ASB				**60-day ASB**			
Percent Volatilized		13.0	%	Percent Volatilized		0.2	%
Mass Volatilized		1.67	lb/ODTUP	Mass Volatilized		0.02	lb/ODTUP
Percent Biodegraded		-5.8	%	Percent Biodegraded		98.4	%
Mass Biodegraded		-0.74	lb/ODTUP	Mass Biodegraded		11.71	lb/ODTUP
Mass Breakthrough		11.90	lb/ODTUP	Mass Breakthrough		0.17	lb/ODTUP

Notes:
The r-factor is updated once a year in the first quarter.

Data Date: 1/31/26

I. BIOTREATMENT UNIT DESCRIPTION

	Units	5-Day ASB Zone 1	5-Day ASB Zone 2	5-Day ASB Zone 3	60-Day ASB Zone 4
Number of 75 HP Surface Aerators	#	0	8	27	23
Total Horsepower	hp	0	600	2025	1725
Temperature	°C	33.3	31.1	33.3	15.6
Length	ft	865	206	1674	2590
Width	ft	432	413	558	1295
Average Depth	ft	14	14	14	6
Aerator Rotation	rpm	1200	1200	1200	1200
Agitation Area per Aerator	ft^2	1452	1452	1452	1452
Impeller Diameter	in	23.9	23.9	23.9	23.9

28.3

II. OVERALL PARAMS - Individual Flows & MeOH Concentrations

Parameter		Flow MGD	MeOH mg/L	MeOH lb/day
5-Day ASB Inlet**		40.50	10.7	3,614
Hard Pipe Inlet		0.41	3,330	11,510

***except condensate flow*

II. OVERALL PARAMS - Total Flows & Wind Speed

	Flow m^3/sec	Flow MGD	MeOH mg/L
5-Day ASB Total	NA	NA	NA
Wind Speed	mph	5.44	

III. HAP DATA - Methanol & Zone Temperatures

	Units	Inlet	5-Day ASB Zone 1	5-Day ASB Zone 2	5-Day ASB Zone 3	Cross-over	60-Day ASB Zone 4	Curtain tip west side
Conc.	mg/L	44.3	33.80	45.70	43.20	41.10	0.57	0.57
Temp.	°F		92.0	88.0	92.0	92	60.0	Detection Limit
								0.4

IV. RESULTS

Partitioning Split by Ponds		5 Day ASB	60 Day ASB
Fraction biodegraded		-5.8	98.4
Fraction air emissions		13.0	0.2
Fraction remaining in unit effluent		92.7	1.4

IV. RESULTS (Net Combined Efficiency of 5-Day & 60-Day ASBs

Overall Partitioning		%
Fraction biodegraded		85.5
Fraction air emissions		13.2
Fraction remaining in unit effluent		1.3

I. ADDITIONAL BIOTREATMENT UNIT DESCRIPTION

Active Volume

5-Day ASB Zone 1	5-Day ASB Zone 2	5-Day ASB Zone 3	60-Day ASB Zone 4		Totals	Units
148,157	33,732	370,349	569,920		1,122,157	m^3
39.1	8.9	97.8	150.6		296.4	MG

Surface Area

5-Day ASB Zone 1	5-Day ASB Zone 2	5-Day ASB Zone 3	60-Day ASB Zone 4		Totals	
373,680	85,078	934,092	3,354,050		4,746,900	ft^2
34,716	7,904	86,780	311,601		441,001	m^2
8.6	2.0	21.4	77.0		109.0	acres

Parameters For Calculating Mass Transfer Coefficients
International Paper - Riverdale Mill

Data Date: 01/31/26

HAP Compound	Diff in Water	Diff in Air	Henry's Law	Equil. Ratio (Hc) or (Keq)	MW	ScG	Antoine Eqtn	
	cm2/s	cm2/s	atm-m3/mol	m3 liq to m3 gas	g/mol		b	c
Methanol	1.64E-05	0.150	5.89E-06	2.41E-04	32.0	1.006	1474	229.13
Acetaldehyde	1.41E-05	0.124	8.77E-05	3.58E-03	45.1	1.216	1600	291.8
MEK	9.80E-06	0.081	1.30E-04	5.31E-03	72.1	1.867	1305	229.27

General Parameters & Constants

PARAMETER	Units	Value	Name
viscosity of air	g/cm-s	0.000181	va
viscosity of water	g/cm-s	0.002	vw
density of air	g/cm3	0.0012	da
density of water	g/cm3	1	dw
MW of air	g/mol	29	Mwa
MW of water	g/mol	18	MWw
Diff of O2 in H2O	cm2/s	2.40E-05	DO2w
grav const.	lb-ft/s2/lb	32.17	g
R	atm-m3/mol K	8.21E-05	R
Aerator Motor Eff	fraction	0.85	AerEff
O2 Trans Correct		0.83	Beta
Wind Speed	m/s	2.43	U
Diff of Ether	m/s	8.50E-06	Dether
O2 Trans	lb O2/HP-h	3	J

These Parameters are used when F/D < 14 AND U > 3.25 m/s

TURBULENT

KL Params	5-Day ASB Zone 1	5-Day ASB Zone 2	5-Day ASB Zone 3	60-Day ASB Zone 4
w	126.3	126.3	126.3	126.3
Re	3.11E+06	3.11E+06	3.11E+06	3.11E+06
PI	0	35,063	35,063	35,063
Power Number, p	0.00E+00	2.86E-04	2.86E-04	2.86E-04
Fr	9.88E+02	9.88E+02	9.88E+02	9.88E+02
Total TurbArea (ft2)	0	3872	13068	33396
Frac. Agitated	0.000	0.046	0.014	0.010
(by surface aerators)				
QUIESCENT				
Depth	4.27	4.27	4.27	1.83
SurfArea (ft2)	373,680	85,078	934,092	3,354,050
SurfArea (m2)	34,762	7,914	86,894	312,011
F/D Ratio	49	23	78	344
ScL - Methanol	NA	NA	NA	NA
ScL - Acetaldehyde	NA	NA	NA	NA
ScL - MEK	NA	NA	NA	NA
U* (Friction Velocity)	NA	NA	NA	NA
DIFFUSED				
Air flow, cfm	0	12,500	12,500	0
Air flow, m3/s	0.000	5.899	5.899	0.000

Calculating Mass Transfer Coefficient KL for Various Zones
International Paper - Riverdale Mill

Data Date: **01/31/26**

Surface Aeration												
	Turbulent Area				Quiescient Area KL, m/s							OVERALL
ZONE	Temp Adj	kG	kL	KL turb	kG		U10 > 3.25			kL	KL quisc	KL overall
	H	m/s	m/s	m/s	m/s	U10 < 3.25	F/D<14	14<F/D<51.2	F/D>51.2	m/s	m/s	m/s
5-Day ASB: Zone1												
Methanol	8.99E-06	0.00E+00	0.00E+00	0.00E+00	5.33E-03	4.31E-06	NA	2.35E-06	2.39E-06	4.31E-06	1.32E-06	1.32E-06
5-Day ASB: Zone2												
Methanol	8.05E-06	1.44E-01	6.14E-02	4.64E-05	5.78E-03	4.31E-06	NA	1.73E-06	2.39E-06	4.31E-06	1.30E-06	3.36E-06
5-Day ASB: Zone3												
Methanol	8.99E-06	1.44E-01	6.47E-02	5.15E-05	5.07E-03	4.31E-06	NA	1.17E-06	2.39E-06	4.31E-06	1.28E-06	1.98E-06
60-Day ASB: Zone4												
Methanol	3.52E-06	1.44E-01	1.42E-02	2.14E-05	4.73E-03	4.31E-06	NA	9.39E-06	2.39E-06	4.31E-06	6.03E-07	8.10E-07

	Submerged Aeration - KD		
	Temp. Adj.	Equiv. KL	KL overall
	Hc (Keq)	m/s	
5-Day ASB: Zone1			
Methanol	3.67E-04	0.00E+00	1.32E-06
5-Day ASB: Zone2			
Methanol	3.29E-04	2.45E-07	3.60E-06
5-Day ASB: Zone3			
Methanol	3.67E-04	2.49E-08	2.00E-06

FORM XIII. DATA FORM FOR THE ESTIMATION OF MULTIPLE ZONE BIODEGRADATION FROM UNIT CONCENTRATIONS

Data Date: **01/31/26**

NAME OF THE FACILITY	**IP - RIVERDALE**	
COMPOUND for site specific biorate determination	**MeOH**	
Number of zones in the biological treatment unit	1	
VOLUME of full-scale system (cubic meters)	2	718076.8224
Average DEPTH of the full-scale system (meters)	3	3.66
FLOW RATE of wastewater to the unit (m3/s)	4	1.774
FLOW RATE of condensate to the unit (m3/s)	5	0.018
Total wastewater flowrate - (including condensates) (m3/s)	5-A	NA
ESTIMATE OF KL (m/s)	6	*see table*
Concentration in the wastewater treated in the unit (mg/L)	7	10.7
Concentration in the condensates (mg/L)	8	3,330
Concentration in wastewater (total - inc. cond) in (mg/L)	8-A	NA
Concentration in the treated effluent (mg/L)	9	0.57
TOTAL INLET FLOW (m3/s) line 4 plus the number on line 5 (or 5-A)	10	1.792
TOTAL RESIDENCE TIME (s) line 2 divided by line 10.	11	400617
TOTAL AREA OF IMPOUNDMENT (m2) line 2 divided by line 3	12	196196

	Lines 13 through 15 Not Used						**Overall**
				Estimate of KL			
Zone	**Concentration for zone,**		**Area of the**	**in the zone**		**AIR STRIPPING**	
Number	**Ci (mg/L)**		**zone, A (m2)**	**(m/s)**		**KL A Ci (g/s)**	
5-Day ASB Z-1	33.80		34714.87	1.32E-06			1.55
5-Day ASB Z-2	45.70		7903.75	3.60E-06			1.30
5-Day ASB Z-3	43.20		86777.15	2.00E-06			7.51
60-Day ASB-Z4	0.57		311591.25	8.10E-07			0.14
TOTALS - sum for each zone.		**15**	**440987.01**		**16**		**10.50**

Removal by air stripping (g/s). Line 16.	17	10.50
Loading in effluent (g/s). Line 9 times line 10.	18	1.02
Total loading (g/s). {(line 5*line 8)+(line 4*line 7)} or {line 5-A*line 8-A}.	19	79.4
Removal by biodegradation (g/s) Line 19 minus (line 17 + line 18).	20	67.9
Fraction biodegraded: Divide line 20 by line 19.	21	0.855
Fraction air emissions: Divide line 17 by line 19.	22	0.132
Fraction remaining in unit effluent. Divide line 18 by 19.	23	0.013

Date	January 31, 2026
Total Production ODTUP/15 day	14,098
Methanol Mass in Hard Pipe (lb/15 day)	180,908
Methanol in condensate (lb/ODUTP):	12.8
Fraction of Methanol Biodegraded Fbio:	0.855
r-factor	0.0230
Total HAPs treated (lb/ODTUP):	**10.7**
sCOD/hp (5-day rolling avg)	**103**

Averages

Fraction of Methanol Biodegraded Fbio:	**0.855**
5-day average Total HAPS Treated	**10.70**
sCOD/hp (lb/hp)	**103**

DATA FORM FOR THE ESTIMATION OF THE HENRY'S LAW CONSTANT FOR A COMPOUND IN THE BIOLOGICAL TREATMENT UNIT			
Form IX			
NAME OF THE FACILITY for site specific biorate determination	Riverdale Mill		
COMPOUND for site specific biorate determination			Methanol
LISTED HENRY'S LAW VALUE AT 25 degrees Celsius. (ratio of mol fraction in gas to mole fraction in water at one atmosphere)	1		0.2885
TEMPERATURE of the liquid in the unit (deg.C)	2		28.333333
CALCULATION OF K			
Temperature adjusted Henry's law value (equals the value on line 1 if the temperature on line 2 is 25)	3		0.3270
Temperature in degrees Kelvin. Add 273.16 to the number on line 2. Enter the results here.	4		301.5
Temperature ratio. Divide 273.16 by the number on line 4. Enter the results here.	5		0.9060
Henry's Law adjustment factor. Multiply the number on line 5 by 0.804 and enter the results here.	6		0.7284
Henry's Law value (g/m3 gas per g/m3 liquid) Multiply the number on line 3 by the number on line 6 and divide the results by 1000. Enter the results here.	7		2.38E-04
Henry's Law value (atm m3 per mol) Divide the number on line 3 by 55555 and enter the results here.	8		5.885E-06

Appendix C

Weston Solutions Report

International Paper - Riverdale Lab Test Results

Proficy Data

Weston Solutions, Inc.
1625 Pumphrey Avenue
Auburn, Alabama 36832-4303
334-466-5600 - Fax 334-466-5660
www.westonsolutions.com

3 February 2026

Carla Pearson
International Paper - Riverdale Mill
601 County Road 78
Selma, AL 36703

Re: Methanol/HAPs Analysis Results

Job #: 2026-062

Dear Ms. Pearson:

This letter with analytical results constitutes our report for the analysis of the samples submitted to our laboratory for methanol and HAPs. The samples were received on 28 January through 2 Feburary 2026 in good condition and in accordance with the Chains-of-Custody. Samples were prepared for analysis on 28 January through 2 Feburary 2026 and analyzed on 28 January through 2 Feburary 2026. The sample preparation and analyses were performed according to NCASI Method DI/MeOH-94.03 for methanol and NCASI Method DI/HAPs-99.01 for HAPs. The HAPs analysis included the following additional anaytes: acetaldehyde, propional, and MEK. The results have been summarized and are presented below in Table 1 and Table 2 of this letter.

RESULTS SUMMARY:

Table 1
Summary of Methanol Results

Source ID	Methanol Concentration (µg/mL)				
	Date Collected				
	1/28/2026	1/29/2026	1/30/2026	1/31/2026	2/1/2026
HP- Hardpipe	3000	3270	3480	3140	3330
IN- Inlet (5 Day)	38.6	34.5	28.6	45.7	10.7
Z1- Zone 1	44.7	32.6	49.1	47.6	33.8
Z2- Zone 2	56.5	41.4	42.7	44.6	45.7
Z3- Zone 3	54.4	43.3	38.5	44.3	43.2
SO- Point 7	49.9	42.3	39.3	39.2	41.1
Z4- Point 10	< 1.14	< 1.14	< 1.14	< 1.14	< 1.14
TC- Point 12	< 1.14	< 1.14	< 1.14	< 1.14	< 1.14

Table 2
Summary of HAPs Results for QPT Hardpipe Composite

Source ID	Date Collected	Acetaldehyde Concentration (µg/mL)	Propional Concentration (µg/mL)	MEK Concentration (µg/mL)
HP- Hardpipe	1/28/2026	43.2	1.81	26.7
HP- Hardpipe	1/29/2026	47.9	< 1.24	27.5
HP- Hardpipe	1/30/2026	42.8	2.02	31.0
HP- Hardpipe	1/31/2026	36.8	1.80	34.9
HP- Hardpipe	2/1/2026	36.6	1.93	38.0

Weston Solutions, Inc.
1625 Pumphrey Avenue
Auburn, Alabama 36832-4303
334-466-5600 - Fax 334-466-5660
www.westonsolutions.com

Ms. Carla Pearson
International Paper

Job #: 2026-062
3 February 2026

CASE NARRATIVE:

Enclosed are copies of the Chains-of-Custody records, acknowledging receipt of these samples. Please note that any unused portion of the samples will be discarded 90 days after the date of receipt.

These results of this report relate only to the samples listed in the body of this report.

This report shall not be reproduced by any organization outside of Weston Solutions, Inc. in part or in full, without the written approval from Weston Solutions, Inc.

These results meet all requirements of NELAP. LELAP Accreditation ID - 03024.

QUALITY ASSURANCE AND QUALITY CONTROL:

Quality control procedures conformed to the requirements of the referenced method and our quality assurance program.

All quality control results associated with this sample set were within acceptable limits and/or do not adversely affect the reported results. The quality control analysis results are shown in the Quality Control section of this report.

We appreciate the opportunity to work with you in performing this analysis. If we can be of any other assistance, please contact me at (334) 399-1580.

Sincerely,
WESTON SOLUTIONS, INC.
Staci Hickman

Staci Hickman
Laboratory Manager

1625 Pumphrey Ave. Auburn, AL 36832
334.466.5600

Analysis Report

for Methanol by NCASI DI/MeOH 94.03

Client : International Paper - Riverdale Mill
Weston W.O. # : 05116.002.031
Dates Received : 1/28/2026-2/2/2026
Date Prepped : 1/28/2026-2/2/2026
Limit of Quantification (µg/mL): 1.14

Lab Job #: 2026-062

Analytical Method: NCASI DI/MeOH-94.03
Instrument ID: GC/FID-Ole Blue
Date(s) Analyzed: 1/28/2026-2/2/2026
Analyst: SH/MB

Source ID	Date Collected	Collection Time	Collection Temperature (F)	Lab ID	Methanol Concentration (µg/mL)
HP- Hardpipe	1/28/2026	7:21 AM	48	BD 4789	**3000**
IN- Inlet (5 Day)	1/28/2026	8:15 AM	98	BD 4790	**38.6**
Z1- Zone 1	1/28/2026	9:15 AM	98	BD 4791	**44.7**
Z2- Zone 2	1/28/2026	9:20 AM	96	BD 4792	**56.5**
Z3- Zone 3	1/28/2026	9:30 AM	94	BD 4793	**54.4**
5O- Point 7	1/28/2026	8:40 AM	94	BD 4794	**49.9**
Z4- Point 10	1/28/2026	9:25 AM	64	BD 4795	**< 1.14**
TC- Point 12	1/28/2026	7:30 AM	58	BD 4796	**< 1.14**
HP- Hardpipe	1/29/2026	4:35 PM	68	BD 4859	**3270**
IN- Inlet (5 Day)	1/29/2026	7:50 AM	88	BD 4804	**34.5**
Z1- Zone 1	1/29/2026	9:15 AM	96	BD 4805	**32.6**
Z2- Zone 2	1/29/2026	9:10 AM	96	BD 4806	**41.4**
Z3- Zone 3	1/29/2026	9:05 AM	96	BD 4807	**43.3**
5O- Point 7	1/29/2026	8:50 AM	96	BD 4808	**42.3**
Z4- Point 10	1/29/2026	8:25 AM	60	BD 4809	**< 1.14**
TC- Point 12	1/29/2026	7:30 AM	60	BD 4810	**< 1.14**
HP- Hardpipe	1/30/2026	7:38 AM	90	BD 4851	**3480**
IN- Inlet (5 Day)	1/30/2026	7:20 AM	90	BD 4852	**28.6**
Z1- Zone 1	1/30/2026	8:05 AM	92	BD 4853	**49.1**
Z2- Zone 2	1/30/2026	8:00 AM	92	BD 4854	**42.7**
Z3- Zone 3	1/30/2026	7:50 AM	94	BD 4855	**38.5**
5O- Point 7	1/30/2026	7:45 AM	92	BD 4856	**39.3**
Z4- Point 10	1/30/2026	7:55 AM	58	BD 4857	**< 1.14**
TC- Point 12	1/30/2026	8:50 AM	62	BD 4858	**< 1.14**
HP- Hardpipe	1/31/2026	7:39 AM	76	BD 4864	**3140**
IN- Inlet (5 Day)	1/31/2026	8:00 AM	88	BD 4865	**45.7**
Z1- Zone 1	1/31/2026	8:15 AM	94	BD 4866	**47.6**
Z2- Zone 2	1/31/2026	8:20 AM	94	BD 4867	**44.6**
Z3- Zone 3	1/31/2026	8:30 AM	92	BD 4868	**44.3**
5O- Point 7	1/31/2026	8:35 AM	86	BD 4869	**39.2**
Z4- Point 10	1/31/2026	8:25 AM	54	BD 4870	**< 1.14**
TC- Point 12	1/31/2026	9:45 AM	62	BD 4871	**< 1.14**
HP- Hardpipe	2/1/2026	5:40 AM	88	BD 4895	**3330**
IN- Inlet (5 Day)	2/1/2026	10:30 AM	76	BD 4878	**10.7**
Z1- Zone 1	2/1/2026	7:50 AM	92	BD 4879	**33.8**
Z2- Zone 2	2/1/2026	7:55 AM	88	BD 4880	**45.7**
Z3- Zone 3	2/1/2026	8:05 AM	92	BD 4881	**43.2**
5O- Point 7	2/1/2026	8:15 AM	92	BD 4882	**41.1**
Z4- Point 10	2/1/2026	8:00 AM	60	BD 4883	**< 1.14**
TC- Point 12	2/1/2026	9:00 AM	62	BD 4884	**< 1.14**

Staci Hickman

Staci Hickman, Laboratory Manager

1625 Pumphrey Ave. Auburn, AL 36832
334.466.5600

Analysis Report

for HAPS by NCASI DI/HAPs 99.01

Client : International Paper - Riverdale Mill
Client W.O. # : 05116.002.031
Date Received : 1/28/2026-2/2/2026
Lab Job #: : 2026-062
Limit of Quantification for Acetaldehyde (µg/mL): 1.20
Limit of Quantification for Propional (µg/mL): 1.24
Limit of Quantification for MEK (µg/mL): 1.63

Instrument ID: GC/FID-Ole Blue
Analyst: SH/MB
Date(s) Prepped: 1/28/2026-2/2/2026
Date(s) Analyzed: 1/28/2026-2/2/2026

Source ID	Date Collected	Sample Dilution	Laboratory ID	Acetaldehyde Concentration (µg/mL)	Propional Concentration (µg/mL)	MEK Concentration (µg/mL)
HP- Hardpipe	1/28/2026	1	BD 4789	43.2	1.81	26.7
HP- Hardpipe	1/29/2026	1	BD 4859	47.9	<1.24	27.5
HP- Hardpipe	1/30/2026	1	BD 4851	42.8	2.02	31.0
HP- Hardpipe	1/31/2026	1	BD 4864	36.8	1.80	34.9
HP- Hardpipe	2/1/2026	1	BD 4895	36.6	1.93	38.0

Staci Hickman

Staci Hickman, Laboratory Manager

WESTON SOLUTIONS
1625 Pumphrey Ave. Auburn, AL 36832
334.466.5600

Quality Control Report
for Methanol by NCASI DI/MeOH 94.03
HAPs by NCASI DI/HAPs 99.01

Client: International Paper - Riverdale Mill
WO#: 05116.002.031

Lab Job #: 2026-062

Analytical Method: NCASI DI/MeOH-94.03 and DI/HAPS-99.01

Table 1.1
Calibration Curve Verification Standards

Analysis Date	Laboratory ID	Methanol		
		Actual Value (µg/mL)	Calculated Value (µg/mL)	Difference (%)
1/28/2026	9340-34-06	54.3	50.6	6.8%
1/28/2026	9340-34-03	1085	1025	5.5%
1/28/2026	LCS 16019400	1005	900	10%
1/28/2026	9340-34-05	109	105	3.7%
1/29/2026	9340-34-06	54.3	50.8	6.4%
1/29/2026	LCS 16019400	1005	906	9.9%
1/29/2026	9340-34-03	1085	1025	5.5%
1/30/2026	LCS 16019400	1005	972	3.2%
1/30/2026	9340-34-05	109	105	3.7%
1/30/2026	9340-34-07	13.6	12.9	5.1%
1/31/2026	9340-34-04	543	516	4.9%
1/31/2026	LCS 16019400	1005	975	3.0%
1/31/2026	9340-34-06	54.3	52.4	3.6%
2/1/2026	9340-34-03	1085	1063	2.0%
2/1/2026	LCS 16019400	1005	978	2.7%
2/1/2026	9340-34-07	13.6	13.3	2.5%
2/2/2026	9340-34-05	109	103	5.5%
2/2/2026	LCS 16019400	1005	981	2.4%
2/2/2026	9340-34-06	54.3	52.5	3.3%

Table 1.2
HAPs Calibration Curve Verification Standards

Analysis Date	Laboratory ID	Acetaldehyde			Propional			MEK		
		Actual Value (µg/mL)	Calculated Value (µg/mL)	Difference (%)	Actual Value (µg/mL)	Calculated Value (µg/mL)	Difference (%)	Actual Value (µg/mL)	Calculated Value (µg/mL)	Difference (%)
1/28/2026	9340-34-06	4.78	4.61	*3.5%*	4.94	4.73	*4.3%*	6.54	6.21	*5.1%*
1/28/2026	9340-34-03	95.7	92.4	*3.5%*	98.9	101	*2.6%*	131	133	*1.3%*
1/28/2026	LCS 16019400	20.0	21.1	*5.6%*	19.9	21.4	*7.5%*	20.0	21.9	*9.4%*
1/28/2026	9340-34-05	9.57	9.44	*1.4%*	9.89	9.70	*2.0%*	13.1	12.7	*3.0%*
1/29/2026	9340-34-06	4.78	4.60	*3.8%*	4.94	4.81	*2.6%*	6.54	6.10	*6.8%*
1/29/2026	LCS 16019400	20.0	21.2	*5.9%*	19.9	21.6	*8.5%*	20.0	21.4	*7.1%*
1/29/2026	9340-34-03	95.7	87.6	*8.5%*	98.9	91.4	*7.6%*	131	114	*13%*
1/30/2026	LCS 16019400	20.0	21.4	*6.8%*	19.9	20.2	*1.6%*	20.0	19.3	*3.7%*
1/30/2026	9340-34-05	9.57	8.86	*7.5%*	9.89	9.21	*6.9%*	13.1	12.0	*8.4%*
1/30/2026	9340-34-07	1.20	1.25	*4.2%*	1.24	1.28	*3.4%*	1.63	1.60	*1.8%*
1/31/2026	9340-34-04	47.8	46.5	*2.6%*	49.4	50.9	*3.1%*	65.4	66.8	*2.2%*
1/31/2026	LCS 16019400	20.0	21.4	*6.9%*	19.9	19.1	*4.0%*	20.0	18.8	*6.2%*
1/31/2026	9340-34-06	4.78	4.88	*2.1%*	4.94	5.03	*1.9%*	6.54	6.50	*0.6%*
2/1/2026	9340-34-03	95.7	96.6	*0.9%*	98.9	103	*4.0%*	131	137	*4.9%*
2/1/2026	LCS 16019400	20.0	19.1	*4.6%*	19.9	19.3	*2.8%*	20.0	18.7	*6.4%*
2/1/2026	9340-34-07	1.20	1.29	*7.6%*	1.24	1.19	*4.2%*	1.63	1.66	*2.1%*
2/2/2026	9340-34-05	9.57	9.04	*5.5%*	9.89	9.81	*0.8%*	13.1	12.9	*1.6%*
2/2/2026	LCS 16019400	20.0	19.8	*1.2%*	19.9	19.3	*3.0%*	20.0	18.8	*6.3%*
2/2/2026	9340-34-06	4.78	4.63	*3.2%*	4.94	4.88	*1.3%*	6.54	6.46	*1.3%*

Staci Hickman
Staci Hickman, Laboratory Manager

1625 Pumphrey Ave. Auburn, AL 36832
334.466.5600

Quality Control Report
for Methanol by NCASI DI/MeOH 94.03
HAPs by NCASI DI/HAPs 99.01

Client: International Paper - Riverdale Mill
WO#: 05116.002.031

Lab Job #: 2026-062

Analytical Method: NCASI DI/MeOH-94.03 and DI/HAPS-99.01

Table 2.1
Replicate Analysis

Analysis Date	Laboratory ID	Methanol		
		Original Value (µg/mL)	Replicate Value (µg/mL)	Difference (%)
1/28/2026	BD 4789	3004	3006	0.0%
1/29/2026	BD 4805	32.6	32.6	0.0%
1/30/2026	BD 4852	28.6	28.4	0.4%
1/31/2026	BD 4870	0.00	0.00	N/A

Analysis
HAPs Replicate Analysis

Analysis Date	Laboratory ID	Acetaldehyde			Propional			MEK		
		Original Value (µg/mL)	Replicate Value (µg/mL)	Difference (%)	Original Value (µg/mL)	Replicate Value (µg/mL)	Difference (%)	Original Value (µg/mL)	Replicate Value (µg/mL)	Difference (%)
1/28/2026	BD 4789	43.2	39.4	*4.6%*	1.81	1.95	*3.9%*	26.7	26.9	*0.4%*

Table 3.1
Methanol Duplicate Analysis

Analysis Date	Laboratory ID	Methanol		
		Original Value (µg/mL)	Duplicate Value (µg/mL)	Difference (%)
1/28/2026	BD 4790	38.6	38.9	0.5%
1/29/2026	BD 4806	41.4	42.4	1.2%
1/30/2026	BD 4851	3476	3618	2.0%
1/31/2026	BD 4871	0.00	0.00	N/A

Table 3.2
HAPs Duplicate Analysis

Analysis Date	Laboratory ID	Acetaldehyde			Propional			MEK		
		Original Value (µg/mL)	Duplicate Value (µg/mL)	Difference (%)	Original Value (µg/mL)	Duplicate Value (µg/mL)	Difference (%)	Original Value (µg/mL)	Duplicate Value (µg/mL)	Difference (%)
1/30/2026	BD 4851	42.8	43.1	*0.4%*	2.02	2.20	*4.4%*	31.0	30.9	*0.2%*

Table 4.1
Spike Analysis

Analysis Date	Laboratory ID	Methanol				
		Original Value (µg/mL)	Spiked Value (µg/mL)	Recovered Amount (µg/mL)	Spiked Amount (µg/mL)	Recovery (%)
1/28/2026	BD 4790	38.6	118	79.9	76.7	104%
1/29/2026	BD 4806	41.4	117	75.5	76.7	98%
1/30/2026	BD 4851	3476	5679	2202	2373	93%
1/31/2026	BD 4871	0.00	8.15	8.15	7.67	106%

Staci Hickman 6
Staci Hickman, Laboratory Manager

1625 Pumphrey Ave. Auburn, AL 36832
334.466.5600

Quality Control Report
for Methanol by NCASI DI/MeOH 94.03
HAPs by NCASI DI/HAPs 99.01

Client: International Paper - Riverdale Mill
WO#: 05116.002.031

Lab Job #: 2026-062

Analytical Method: NCASI DI/MeOH-94.03 and DI/HAPS-99.01

Table 4.2
HAPs Spike Analysis

Analysis Date	Laboratory ID	Acetaldehyde				
		Original Value (µg/mL)	Spiked Value (µg/mL)	Recovered Amount (µg/mL)	Spiked Amount (µg/mL)	Recovery (%)
1/30/2026	BD 4851	42.8	87.8	45.0	48.6	*93%*

Analysis Date	Laboratory ID	Propional				
		Original Value (µg/mL)	Spiked Value (µg/mL)	Recovered Amount (µg/mL)	Spiked Amount (µg/mL)	Recovery (%)
1/30/2026	BD 4851	2.02	49.5	47.5	48.5	*98%*

Analysis Date	Laboratory ID	MEK				
		Original Value (µg/mL)	Spiked Value (µg/mL)	Recovered Amount (µg/mL)	Spiked Amount (µg/mL)	Recovery (%)
1/30/2026	BD 4851	31.0	80.9	49.9	47.6	*105%*

- Note the actual analytical result rather than the LOQ was used when the analytical result was less than the LOQ.
- Consequently, certain differences in actual and calculated values may be skewed.

Calculations:

- Standard % Difference = ((|(Actual Value-Calculated Value)|)÷(Actual Amount))*100.
- Replicate/Duplicate % Difference = ((|Average Value - Original Value|) ÷ (Average Value))*100.
- Spike % Recovery = (Recovered Amount) ÷ (Spiked Amount) * 100.

Acceptance Cirteria:

-The CCV Acceptance Criterion is ± 10 % for Methanol. -The Replicate and Duplicate Acceptance Criterion is ± 10 %.
-The CCV Acceptance Criterion is ± 15 % for HAPs -The Spike Recovery Acceptance Criterion is 100 % ± 30 %.
-The LCS Acceptance Criterion is ± 15% for Methanol and ± 20% for HAPs.

Lot #s:

Internal Std 9340-02-00
MeOH Spk Neat 145674 9340-35-01
HAPS Spk 9340-33-01

Staci Hickman

Staci Hickman, Laboratory Manager

CHAIN OF CUSTODY RECORD

Requester: Carla C. Pearson
Requester's: International Paper – Riverdale Mill
601 County Road 78
Selma, Alabama 36703
Phone: 334/418-5236
Fax: 901/473-5470

☒ Samples shipped with ice packs.

Date	Time	Sample Identification	Initials	# of vials	MeOH	HAPS	Other*	Preservative	Sample pH	Sample Description	Temperature Fahrenheit
1/28/26	7:21 Am/Pm	HP--HARDPIPE	HLH	1	X	X		None	30 4789	FOUL CONDENSATE	48°
1/28/26	8:15 Am/Pm	IN – INLET (5Day)	HLH	1	X			HCl	4790	5 DAY INLET	98°
1/28/26	9:15 Am/Pm	Z 1 - ZONE 1	HLH	1	X			HCl	4791	5 Day-- ZONE 1 - POINT 2	98°
1/28/26	9:20 Am/Pm	Z 2 - ZONE 2	HLH	1	X			HCl	4792	5 Day-- ZONE 2 - POINT K	96°
1/28/26	9:30 Am/Pm	Z 3 - ZONE 3	HLH	1	X			HCl	4793	5 Day-- ZONE 3 - POINT R	94°
1/28/26	8:40 Am/Pm	5O--POINT 7	HLH	1	X			HCl	4794	5 DAY OUTLET-- POINT 7	94°
1/28/26	9:25 Am/Pm	Z 4 - POINT 10	HLH	1	X			HCl	4795	60 Day-- ZONE 4 - POINT 10	64°
1/28/26	9:30 Am/Pm	TC – POINT 12	HLH/S	1	X			HCl	4796	60 Day-- TIP OF CURTAIN POINT 12	58°

Relinquished By: Date / Time 1/28/26 10:10
(Signature) Pearson

Relinquished By: Date / Time
(Signature)

Relinquished By: Date / Time
(Signature)

Received By: Date / Time 1/28/26 10:15
(Signature) Custody Seal
for Sayer

Received By: Date / Time
(Signature)

Received By Date / Time
(Signature)

LAB USE ONLY

Received for Laboratory by: Date/Time Custody Intact: Yes ✓ No__ Custody Seal No: Request No: Cooler Temp:
(Signature) Staci Hickman 1/28/26 13:35 2026-062 2.9°C
Laboratory Remarks:

Page 1 of 1

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CHAIN OF CUSTODY RECORD

Requester: Carla C. Pearson
Requester's: International Paper – Riverdale Mill
601 County Road 78
Selma, Alabama 36703
Phone: 334/418-5236
Fax: 901/473-5470

☐Samples shipped with ice packs.

Date	Time	Sample Identification	Initials	# of vials	MeOH	HAPS	Other*	Preservative	Sample pH	Sample Description	Temperature Fahrenheit
1/29/26	9 : 52 Am/Pm	HP--HARDPIPE	JC3	1	X			None	BD	4803 FOUL CONDENSATE	68 °
1/29/26	7 : 50 Am/Pm	IN – INLET (5Day)	EA	1	X			HCl		4804 5 DAY INLET	88 °
1/29/26	9 : 15 Am/Pm	Z 1 - ZONE 1	EA	1	X			HCl		4805 5 Day-- ZONE 1 - POINT 2	96 °
1/29/26	9 : 10 Am/Pm	Z 2 - ZONE 2	EA	1	X			HCl		4806 5 Day-- ZONE 2 - POINT K	96 °
1/29/26	9 : 05 Am/Pm	Z 3 - ZONE 3	EA	1	X			HCl		4807 5 Day-- ZONE 3 - POINT R	96 °
1/29/26	8 : 50 Am/Pm	5O--POINT 7	EA	1	X			HCl		4808 5 DAY OUTLET-- POINT 7	96 °
1/29/26	8 : 25 Am/Pm	Z 4 - POINT 10	EA	1	X			HCl		4809 60 Day-- ZONE 4 - POINT 10	90 °
1/29/26	7 : 30 Am/Pm	TC – POINT 12	KB	1	X			HCl		4810 60 Day-- TIP OF CURTAIN POINT 12	60°F °

Relinquished By: (Signature) Edgar Ly Date / Time 1/29/26 10:00 AM
Relinquished By: (Signature) _____ Date / Time
Relinquished By: (Signature) _____ Date / Time

Received By: (Signature) Edgar Ly Date / Time 1/29/26 10:00 AM
Received By: (Signature) _____ Date / Time
Received By (Signature) _____ Date / Time

LAB USE ONLY

Received for Laboratory by: (Signature) Stan Hickman
Date/Time 1/29/26 12:35
Custody Intact: Yes ✓ No__ Custody Seal No:
Request No: 2026-062
Cooler Temp: 3.3°C
Laboratory Remarks:

Page ____ of ____

CHAIN OF CUSTODY RECORD

Requester: Carla C. Pearson
Requester's: International Paper – Riverdale Mill
601 County Road 78
Selma, Alabama 36703
Phone: 334/418-5236
Fax: 901/473-5470

1/30 26 9:00 AM Polymer Addicts KB 1 X None BD 4859
1/29/26 4:35 PM HP-Hardpipe KB 1 X None BD 4860 for Methane

Samples shipped with ice packs.

Date	Time	Sample Identification	Initials	# of vials	MeOH	HAPS	Other*	Preservative	Sample pH	Sample Description	Temperature Fahrenheit
1/30/26	7:38 Am/Pm	HP--HARDPIPE	JCB	1	X			None	BD 4859	FOUL CONDENSATE	90°
1/30/26	7:20 Am/Pm	IN – INLET (5Day)	EA	1	X			HCl	4860	5 DAY INLET	90°
1/30/26	8:05 Am/Pm	Z 1 - ZONE 1	EA	1	X			HCl	4861	5 Day-- ZONE 1 - POINT 2	92°
1/30/26	8:00 Am/Pm	Z 2 - ZONE 2	EA	1	X			HCl	4862	5 Day-- ZONE 2 - POINT K	92°
1/30/26	7:50 Am/Pm	Z 3 - ZONE 3	EA	1	X			HCl	4863	5 Day-- ZONE 3 - POINT R	94°
1/30/26	7:45 Am/Pm	5O--POINT 7	EA	1	X			HCl	4864	5 DAY OUTLET-- POINT 7	92°
1/30/26	7:55 Am/Pm	Z 4 - POINT 10	EA	1	X			HCl	4865	60 Day-- ZONE 4 - POINT 10	58°
1/30/26	8:50 Am/Pm	TC – POINT 12	KB	1	X			HCl	4866	60 Day-- TIP OF CURTAIN POINT 12	62°

Relinquished By: Date / Time 10:00AM
(Signature) 01/30/2026

Relinquished By: Date / Time
(Signature)

Relinquished By: Date / Time
(Signature)

Received By: Date / Time 10:00AM
(Signature) 01/30/2026

Received By: Date / Time
(Signature)

Received By Date / Time
(Signature)

LAB USE ONLY

Received for Laboratory by: Date/Time Custody Intact: Yes No Custody Seal No: Request No: Cooler Temp:

(Signature) M. Bowman 1/30/26 1225 2026-062 5.3°C

Laboratory Remarks:

Page ____ of ____

10

CHAIN OF CUSTODY RECORD

Requester: Carla C. Pearson
Requester's: International Paper – Riverdale Mill
601 County Road 78
Selma, Alabama 36703
Phone: 334/418-5236
Fax: 901/473-5470

☐Samples shipped with ice packs.

Date	Time	Sample Identification	Initials	# of vials	MeOH	HAPS	Other*	Preservative	Sample pH	Sample Description	Temperature Fahrenheit
1/31/26	7:39 Am/Pm	HP--HARDPIPE	JC3	1	X			None	BD	4864 FOUL CONDENSATE	76 °
1/31/26	8:00 Am/Pm	IN – INLET (5Day)	HLH	1	X			HCl		4865 5 DAY INLET	88 °
1/31/26	8:15 Am/Pm	Z 1 - ZONE 1	HLH	1	X			HCl		4866 5 Day-- ZONE 1 - POINT 2	94 °
1/31/26	8:20 Am/Pm	Z 2 - ZONE 2	HLH	1	X			HCl		4867 5 Day-- ZONE 2 - POINT K	94 °
1/31/26	8:30 Am/Pm	Z 3 - ZONE 3	HLH	1	X			HCl		4868 5 Day-- ZONE 3 - POINT R	92°
1/31/26	8:35 Am/Pm	5O--POINT 7	HLH	1	X			HCl		4869 5 DAY OUTLET-- POINT 7	86 °
1/31/26	8:25 Am/Pm	Z 4 - POINT 10	HLH	1	X			HCl		4870 60 Day-- ZONE 4 - POINT 10	54 °
1/31/26	9:45 Am/Pm	TC – POINT 12	HLH	1	X			HCl		4871 60 Day-- TIP OF CURTAIN POINT 12	68 °

Relinquished By: Date / Time Relinquished By: Date / Time Relinquished By: Date / Time
(Signature) Heidi L Howland 1-31-26 10:00AM (Signature) (Signature)

Received By: Date / Time Received By: Date / Time Received By Date / Time
(Signature) Custody Seal 1-31-26 10:10AM (Signature) (Signature)

LAB USE ONLY

Received for Laboratory by: Date/Time Custody Intact: Yes✓ No__ Custody Seal No: Request No: Cooler Temp:
(Signature) Staci Hickman 1/31/26 12:52 2026-062 3.1°C
Laboratory Remarks:

Page ____ of ____

CHAIN OF CUSTODY RECORD

Requester: Carla C. Pearson
Requester's: International Paper – Riverdale Mill
601 County Road 78
Selma, Alabama 36703
Phone: 334/418-5236
Fax: 901/473-5470

☐Samples shipped with ice packs.

Date	Time	Sample Identification	Initials	# of vials	MeOH	HAPS	Other*	Preservative	Sample pH	Sample Description	Temperature Fahrenheit
2/01/26	: Am / Pm	HP--HARDPIPE		1	X			None		FOUL CONDENSATE	°
2/01/26	10 : 30 Am / Pm	IN – INLET (5Day)	EA	1	X			HCl	BD 4878	5 DAY INLET	76 °
2/01/26	7 : 50 Am / Pm	Z 1 - ZONE 1	EA	1	X			HCl	4879	5 Day-- ZONE 1 - POINT 2	92 °
2/01/26	7 : 55 Am / Pm	Z 2 - ZONE 2	EA	1	X			HCl	4880	5 Day-- ZONE 2 - POINT K	88 °
2/01/26	8 : 05 Am / Pm	Z 3 - ZONE 3	EA	1	X			HCl	4881	5 Day-- ZONE 3 - POINT R	92 °
2/01/26	8 : 15 Am / Pm	5O--POINT 7	EA	1	X			HCl	4882	5 DAY OUTLET-- POINT 7	92 °
2/01/26	8 : 00 Am / Pm	Z 4 - POINT 10	EA	1	X			HCl	4883	60 Day-- ZONE 4 - POINT 10	60 °
2/01/26	9 : 00 Am / Pm	TC -- POINT 12	EA	1	X			HCl	4884	60 Day-- TIP OF CURTAIN POINT 12	62 °

Relinquished By: Date / Time ~~EA 9:45AM~~ 10:30 AM
(Signature) 2/01/2026

Relinquished By: Date / Time
(Signature)

Relinquished By: Date / Time
(Signature)

Received By: Date / Time ~~EA 9:45AM~~ 10:30 AM
(Signature) 2/01/2026

Received By: Date / Time
(Signature)

Received By Date / Time
(Signature)

LAB USE ONLY

Received for Laboratory by: Date/Time Custody Intact: (Yes) No Custody Seal No: Request No: Cooler Temp:

(Signature) M. Bowman 2/1/26 1255 2026-062 5.1°c
Laboratory Remarks:

Page ____ of _____

CHAIN OF CUSTODY RECORD (CONDENSATE COLLECTION TANK)

Weston Solutions, Inc.
1625 Pumphrey Avenue
Auburn, AL 36832
Phone: 334-466-5666

Requested by: Carla C. Pearson
International Paper – Riverdale Mill
601 County Road 78
Selma, Alabama 36703

Phone: 334/418-5232 Fax: 334/418-5115

SAMPLES ARE STORED BETWEEN 1 – 4°C PRIOR TO SHIPMENT

☒ **Samples shipped with ice packs**

Date	Time of collection	Temperature @ time of collection	Initials	# of vials	MeOH	Collect Method	Preservative	Sample pH	Sample Description - Sample Identification	Time sample refrigerated	Temperature @ time of refrigeration	Remarks
02/01/26	5:40Am	88°	EA	1	X	Grab			Cond Collection Tank - CR-RD-MT-IN	6:00Am	78°	BD 4895
02/02/26	9:33Am	96°	EA	1	X	Grab			Cond Collection Tank - CR-RD-MT-IN	9:00Am	88°	
				1	X	Grab			Cond Collection Tank - CR-RD-MT-IN			
				1	X	Grab			Cond Collection Tank - CR-RD-MT-IN			
				1	X	Grab			Cond Collection Tank - CR-RD-MT-IN			
				1	X	Grab			Cond Collection Tank - CR-RD-MT-IN			
				1	X	Grab			Cond Collection Tank - CR-RD-MT-IN			

Relinquished By: (Signature) [signature] Date/Time 02/02/26 9:15Am

Relinquished By: (Signature) _____ Date/Time _____

Relinquished By: (Signature) _____ Date/Time _____

Received By: (Signature) [signature] Date/Time 02/02/26 9:15Am

Received By: (Signature) _____ Date/Time _____

Received By (Signature) _____ Date/Time _____

Page 1 of 1

LAB USE ONLY

Received for Laboratory by: (Signature) M. Bowman Date/Time 2/2/26 1215

Laboratory Remarks:

Custody Intact: (Yes) No__ Custody Seal No:

Request No: 2026-062

Cooler Temp: 5.9°c

END

OF

DOCUMENT

Server-PA	S02CSQLP02A\PRD1												
Start Date	01-28-2026 07:00:00												
End Date	02-01-2026 07:00:00												
Frequency	ANY												
VarID		206	199	198	202	155	1333	158	216	1340	2065	2064	
Source Type		PA	PA	PA	PA	PA	PA	PA	PA	PA	PA	PA	
Calc Formula													
Low Limit													
High Limit													
Eng Units			lbs	ODTP	ODUBTP			Gals	Gals	lbs sCOD/H	lbs sCOD/H	lbs sCOD/HP	
Description	TimeStamp	Mill Day	15 Day - MeOH Collected	15 Day - Digester Tons	Daily - Total Digester Tons	No. Blowers Running	No. Aerators Running Curtained Area	Hard Pipe Influent Flow	Whole Mill Influent Flow (Final)	Total ASB Lbs sCOD/HP w/ Curtain Area	Total ASB Lbs sCOD/HP w/ Curtain Area	5-Day Total ASB LBS sCOD/HP	
		1/27/2026	203,036	15,547	752.6	4	21	297,158	41,899,684	101	101	82	
		1/28/2026	201,254	15,321	746.5	4	21	414,694	39,060,036	95	95	83	
		1/29/2026	192,094	14,700	576.1	4	21	229,549	36,752,284	152	152	100	
		1/30/2026	183,166	14,303	904.8	4	21	384,639	39,965,472	82	82	102	
		1/31/2026	180,908	14,098	891.6	4	21	414,438	40,501,256	85	85	103	

ORIGIN ID:MGMA (334) 418-5236
CARLA PEARSON
INTERNATIONAL PAPER
601 COUNTY ROAD 78

SELMA, AL 36703
UNITED STATES US

SHIP DATE: 10MAR26
ACTWGT: 2.00 LB
CAD: 109152802/INET4535

BILL SENDER

TO **MR. STEVEN BISSEY**
ADEM
1400 COLISEUM BLVD
1Q26CVT
MONTGOMERY AL 36130
(334) 271-7700 REF:
INV:
PO: DEPT:

58KJ3/7351/484B

FedEx Express

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MGMA 36130
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